A BILINEAR TWISTED CUBIC MOMENT OF THE RIEMANN ZETA FUNCTION IN THE FIXED-WEIL RANGE $H^{3/2}K \le T^{1-\eta}$

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Abstract. We evaluate a smoothed bilinear twist of a shifted cubic product of the Riemann zeta function. For two independent divisor-bounded coefficient sequences supported on $h \le H$ and $k \le K$, we prove the expected three-term shifted main term, uniformly for shifts of size $O((\log T)^{-1})$, with error

$$O_{\eta,C,\varepsilon}\Big(T^{\varepsilon}\{KH^{3/2} + T^{1/2}(HK)^{1/2}\}\Big)$$

whenever $H^{3/2}K \le T^{1-\eta}$. The proof fixes one Poisson orientation throughout. Its nonzero frequencies are controlled by finite completion and the Weil bound after exact extraction of (h, km). The zero frequency is evaluated with a coupled two-variable smoothing. A full-kernel regulator and genuine branchwise Tonelli domains justify the beta integrals; the diagonal and zero-mode residues are assembled before analytic continuation or the final regulator move. A local three-divisor argument proves joint holomorphy when the shifts collide.

Contents

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1. Introduction

Moments of the Riemann zeta function twisted by Dirichlet polynomials are a basic input to mollification and to the study of zeros on the critical line. The twisted second moment is classical [1], while the fourth moment has been evaluated in substantial polynomial-length ranges [6, 2]. The present paper concerns a three-zeta analogue of the twisted second and fourth moments. For smooth w concentrated at height T, consider

$$\sum_{h\le H}\sum_{k\le K}\frac{a_h\overline{b_k}}{\sqrt{hk}}\int_{\mathbb{R}} w(t)\zeta\Big(\tfrac12+\alpha+it\Big)\,\zeta\Big(\tfrac12+\beta+it\Big)\,\zeta\Big(\tfrac12+\gamma-it\Big)\left(\frac{h}{k}\right)^{it}\,\mathrm{d}t. \tag{1.1}$$

Here a_h and b_k are independent divisor-bounded sequences and the three shifts are $O((\log T)^{-1})$.

Bui introduced the corresponding twisted cubic moment in connection with a three-piece mollifier [4]. The formula in that note was left unproved, and its mollifier application uses two arithmetically different coefficient sequences. We therefore formulate the bilinear problem with independent a_h and b_k. Our main result proves the expected three permutation terms in the fixed-Weil range

$$H^{3/2}K\le T^{1-\eta}. \tag{1.2}$$

The exact statement, including a uniform error term and the interpretation at colliding shifts, is Theorem 2.1. Its three terms agree with the shifted moment recipe of Conrey, Farmer, Keating, Rubinstein and Snaith [5].

The closest established technology includes the twisted fourth moment of Hughes and Young [6], the quadratic divisor theorem of Bettin, Bui, Li and Radziwill [2], and the twisted second-moment work of Bettin, Chandee and Radziwill [3]. Those theorems have different zeta-factor or coefficient structures and do not specialize to (1.1). In particular, polarization of a quadratic form with a common structured factor does not create two arbitrary coefficient sequences in the cubic integral. The proof below instead retains the degenerate equation

$$hn - klm = r$$

coming from the cubic approximate functional equation.

Pliego's asymptotic formulas for special mixed cubic moments [7] involve a fixed one-sided polynomial and likewise do not contain the two arbitrary, asymmetric coefficient sequences in (1.1).

There are two independent analytic issues. For $r \neq 0$, Poisson summation must be carried out only after the common divisor $d = (h, km)$ is extracted; otherwise the modular inverse is not defined. Keeping a single l-Poisson orientation, finite completion and the Weil bound give

$$\mathcal{R}_{\neq 0}^{(l)} \ll_\varepsilon T^\varepsilon K H^{3/2}.$$

For the zero frequency, formal deletion of the continuous endpoint regions $0 < l < 1$ and $0 < n < 1$ leads to beta integrals outside a common absolute convergence chamber. We resolve this with a coupled (s, u) kernel. Finite variable and Abel regulators are kept while finite contour rectangles are deformed; the rectangle heights tend to infinity before either regulator is removed. Thus the source limit is controlled by the full two-variable kernel, whereas the destination limit is taken in genuine Tonelli domains.

The diagonal and all four zero-mode branches must then be assembled at fixed u. Three pairs of small s-residues cancel, including their finite Euler factors and exact gamma factors. Quantitative holomorphic continuation is applied only to the complete endpoint error after subtracting the full diagonal boundary term. The u-contours are moved afterward. This ordering is what makes the result uniform at $\alpha = \beta = \gamma = 0$.

If $H \leq T^{\theta_1}$ and $K \leq T^{\theta_3}$, the proved open region is

$$\frac{3}{2}\theta_1 + \theta_3 < 1.$$

It includes Bui's boxes $H \leq T^{4/7-\varepsilon_B}$, $K \leq T^{1/4-\varepsilon_B}$ whenever $\varepsilon_B > 3/70$. It does not include the limiting corner $(4/7, 1/4)$; at that corner the pointwise Weil budget is $T^{31/28+o(1)}$. We discuss this exact boundary, without using it in the proof, in Section 10.

Organization. Section 2 states the theorem and records the Euler factors. Section 3 constructs the coupled smoothing and proves its full-contour bounds. Section 4 gives the exact gcd and Poisson decomposition, and Section 5 proves the nonzero-frequency estimate. The endpoint deletion and regulator bridge occupy Section 6. The beta, Euler and archimedean identities are established in Section 7. The fixed-u residue assembly and quantitative continuation are performed in Section 8. The final u-moves and the collision argument appear in Section 9. An audit and reproducibility ledger is given in Appendix A.

2. Statement of the theorem

Let $C, \eta > 0$ be fixed and let T be sufficiently large in terms of these parameters. Put

$$\Delta = \frac{T}{\log T}.$$

Throughout, $w \in C_c^\infty(\mathbb{R})$ is supported in $[T/4, 2T]$ and satisfies

$$w^{(j)}(t) \ll_j \Delta^{-j} \qquad (j \geq 0). \tag{2.1}$$

We make the small-power and seminorm convention explicit. For $r \geq 0$ put

$$\|w\|_{r,\Delta} := \max{}_{0\leq j\leq r} \Delta^j \|w^{(j)}\|_\infty, \tag{2.2}$$

$$\|(U, \boldsymbol{\omega})\|_r := \max{}_{0\leq j\leq r} \left\{ \|U^{(j)}\|_\infty, \max{}_{\omega\in\boldsymbol{\omega}} \|\omega^{(j)}\|_\infty \right\}, \tag{2.3}$$

where U is the endpoint cutoff introduced in Section 4 and $\boldsymbol{\omega}$ denotes the five fixed dyadic cutoffs used in Section 5. These auxiliary cutoffs are fixed once and are not additional data in the theorem.

Whenever T^ε or $\ll_\varepsilon$ occurs, $\varepsilon > 0$ denotes the final prescribed small exponent. Auxiliary divisor, localization, and dyadic exponents are chosen sufficiently smaller in terms of this ε, after the finite derivative orders have been fixed. The implied constants and the lower threshold for T may depend on η, C, ε, on the fixed smoothing parameters B_0, M, on fixed contour and integration-by-parts orders, on the coefficient-bound constants at finitely many auxiliary exponents, and on (2.2)–(2.3) for some finite r; they are uniform in H, K, T and in the shifts in (2.10). The kernels $\mathcal{G}$ and $\mathscr{V}^\pm$ introduce no further independent seminorms, since their bounds are determined by C, B_0, M and the displayed contours. When an $O_D(T^{-D})$ remainder is requested, M and the later integration-by-parts orders may depend on D as specified in Sections 3 and 8; for Theorem 2.1 a sufficiently large D is fixed in terms of the displayed theorem parameters before T varies. In the final theorem B_0, M, U and $\boldsymbol{\omega}$ are proof-internal fixed choices and are suppressed from the $O_{\eta,C,\varepsilon}$ notation; only finitely many scaled seminorms of the external weight w remain. In particular, a harmless T^ε may absorb $(\log T)^{C_M}$ for fixed M, but never a fixed positive power $T^{\delta(M)}$.

For positive integers h, k and complex A, B, C_1, define first in a region of absolute convergence

$$Z_{A,B,C_1}(h,k) = \sum_{kab=hc} a^{-1/2-A} b^{-1/2-B} c^{-1/2-C_1}. \tag{2.4}$$

The letters a, b, c in this sum are local summation variables and are unrelated to the coefficient sequences in the theorem. To make the continuation explicit, write $g = (h,k)$, $h = gh_0$, $k = gk_0$, and set

$$q_Z = 1 + A + C_1, \qquad v_Z = 1 + B + C_1.$$

Then

$$Z_{A,B,C_1}(h,k) = h_0^{1/2+C_1} k_0^{-1/2-C_1} \zeta(q_Z)\zeta(v_Z)$$

$$\times \prod_{p^\nu \| h_0} (1-p^{-qz})(1-p^{-vz}) \sum_{i+j\geq \nu} p^{-qzi-vzj}. \tag{2.5}$$

We use the right side as the meromorphic continuation of (2.4). Its only shift poles are the two displayed zeta poles.

The exact archimedean factor is

$$X_{A,C_1}(t) = \pi^{A+C_1} \frac{\Gamma((\frac{1}{2}-A-it)/2)\Gamma((\frac{1}{2}-C_1+it)/2)}{\Gamma((\frac{1}{2}+A+it)/2)\Gamma((\frac{1}{2}+C_1-it)/2)}. \tag{2.6}$$

For two independent complex sequences (a_h) and (b_k), put

$$\mathcal{I}_{\alpha,\beta,\gamma}(a,b) = \sum_{h\leq H}\sum_{k\leq K} \frac{a_h\overline{b_k}}{\sqrt{hk}} \int_{\mathbb{R}} w(t)\zeta\Big(\tfrac{1}{2}+\alpha+it\Big)\,\zeta\Big(\tfrac{1}{2}+\beta+it\Big)$$

$$\times\, \zeta\Big(\tfrac{1}{2}+\gamma-it\Big)\left(\frac{h}{k}\right)^{it} \mathrm{d}t. \tag{2.7}$$

Theorem 2.1 (Fixed-Weil bilinear twisted cubic moment)**.** *Suppose $H, K \geq 1$ and*

$$H^{3/2}K \leq T^{1-\eta}. \tag{2.8}$$

Let

$$|a_h| \ll_\varepsilon h^\varepsilon, \qquad |b_k| \ll_\varepsilon k^\varepsilon \tag{2.9}$$

for every $\varepsilon > 0$. Uniformly for complex shifts satisfying

$$|\alpha|+|\beta|+|\gamma| \leq \frac{C}{\log T}, \tag{2.10}$$

one has

$$\begin{aligned}\mathcal{I}_{\alpha,\beta,\gamma}(a,b) = \sum_{h\leq H}\sum_{k\leq K} \frac{a_h\overline{b_k}}{\sqrt{hk}} \int_{\mathbb{R}} w(t)\{Z_{\alpha,\beta,\gamma}(h,k)\\ + X_{\alpha,\gamma}(t)Z_{-\gamma,\beta,-\alpha}(h,k) + X_{\beta,\gamma}(t)Z_{\alpha,-\gamma,-\beta}(h,k)\}\,\mathrm{d}t\\ + O_{\eta,C,\varepsilon}\Big(T^\varepsilon\{KH^{3/2}+T^{1/2}(HK)^{1/2}\}\Big).\end{aligned} \tag{2.11}$$

The expression in braces is its jointly holomorphic continuation across

$$\alpha+\gamma=0, \qquad \beta+\gamma=0, \qquad \alpha-\beta=0,$$

including their common collision locus $\alpha=\beta=-\gamma$.

The theorem is formulated with exact gamma ratios. Uniform Stirling asymptotics give

$$X_{A,C_1}(t) = \left(\frac{t}{2\pi}\right)^{-A-C_1}\{1+O_C(T^{-1})\} \tag{2.12}$$

on the support of w. Consequently the two exact factors in (2.11) may be replaced by the corresponding powers used in Bui's formulation. The total replacement error is $O(T^\varepsilon(HK)^{1/2})$.

Corollary 2.2. *If $H \le T^{\theta_1}$ and $K \le T^{\theta_3}$ range over a compact subset of*

$$\frac{3}{2}\theta_1 + \theta_3 < 1,$$

then (2.11) *has a power-saving error. More precisely, under* (2.8)*, choosing the harmless T^{ε} loss with $\varepsilon < \eta/4$ gives an error $O(T^{1-\eta/4})$.*

Remark 2.3. No relation between (a_h) and (b_k) is used. Complex conjugation on b_k is only notational; analytically the theorem is bilinear in two independent sequences.

3. Coupled smoothing and the approximate functional equation

Fix $B_0 > 2$ and put $Y = T^{B_0}$. An integer $M \ge 1$ will be chosen after the required final saving has been fixed. Define

$$R_M(u) = \prod_{j=1}^{M} (1 + u/j) \tag{3.1}$$

and

$$\mathcal{G}(s,u) = e^{s^2} R_M(u) \frac{\big((1-\alpha-\gamma-u)^2 - s^2\big)\big((1-\alpha+\beta-u)^2 - s^2\big)}{(1-\alpha-\gamma)^2(1-\alpha+\beta)^2}, \tag{3.2}$$

$$G(s) = \mathcal{G}(s,0), \qquad Q(u) = \mathcal{G}(0,u). \tag{3.3}$$

Thus G is even, $Q(0) = 1$, and the four s-zeros are

$$s = \pm(1-\alpha-\gamma-u), \qquad s = \pm(1-\alpha+\beta-u). \tag{3.4}$$

The kernel is invariant under $(\beta,\gamma) \mapsto (-\gamma,-\beta)$. In particular, no denominator in (3.2) becomes small in the shift range (2.10).

For shifts B, C_1, set

$$g_{B,C_1}(s,t) = \pi^{-s} \frac{\Gamma\big((\tfrac12 + B + s + it)/2\big)\Gamma\big((\tfrac12 + C_1 + s - it)/2\big)}{\Gamma\big((\tfrac12 + B + it)/2\big)\Gamma\big((\tfrac12 + C_1 - it)/2\big)}. \tag{3.5}$$

The factor π^{-s} is included here, so the Mellin argument below is mn, rather than πmn.

Define

$$V_u^+(x,t) = \frac{1}{2\pi i} \int_{(2)} \frac{\mathcal{G}(s,u)}{s} g_{\beta,\gamma}(s,t) x^{-s} \, \mathrm{d}s, \tag{3.6}$$

$$V_u^-(x,t) = \frac{1}{2\pi i} \int_{(2)} \frac{\mathcal{G}(s,u)}{s} g_{-\gamma,-\beta}(s,t) x^{-s} \, \mathrm{d}s. \tag{3.7}$$

Lemma 3.1 (Unnormalized two-piece approximate functional equation)**.** *For every $A_0 > 0$, uniformly for $t \in \operatorname{supp} w$, for the shifts in* (2.10)*, and for $|\Im u| \le (\log T)^2$, one has*

$$Q(u)\zeta\Big(\tfrac12 + \beta + it\Big)\,\zeta\Big(\tfrac12 + \gamma - it\Big)$$

$$= \sum_{m,n\geq 1} \frac{(n/m)^{it}}{m^{1/2+\beta} n^{1/2+\gamma}} V_u^+(mn,t)$$

$$+ X_{\beta,\gamma}(t) \sum_{m,n\geq 1} \frac{(n/m)^{it}}{m^{1/2-\gamma} n^{1/2-\beta}} V_u^-(mn,t) + O_{A_0}(T^{-A_0}). \tag{3.8}$$

The error remains $O_{A_0}(T^{-A_0})$ after integration against $\Gamma(u)$ on any fixed vertical line.

Proof. On $\Re s = 2$, expand the two zeta functions in

$$\frac{1}{2\pi i} \int_{(2)} \frac{\mathcal{G}(s,u)}{s} g_{\beta,\gamma}(s,t) \zeta\Big(\tfrac{1}{2}+\beta+s+it\Big) \zeta\Big(\tfrac{1}{2}+\gamma+s-it\Big)\, \mathrm{d}s.$$

Move the line to $\Re s = -2$, apply both functional equations, and replace s by $-s$. Evenness of $\mathcal{G}$ produces the dual sum, while the residue at $s=0$ is $Q(u)$ times the zeta product.

There are also four completed-zeta poles,

$$\tfrac{1}{2}-\beta-it, \quad -\tfrac{1}{2}-\beta-it, \quad \tfrac{1}{2}-\gamma+it, \quad -\tfrac{1}{2}-\gamma+it.$$

Their imaginary parts have size T, and $e^{s^2} = \exp(-T^2+O(T))$ there. Their residues and the horizontal integrals are therefore smaller than every power of T. This proves (3.8). Notice that no division by $Q(u)$ has been made. □

The smoothed long zeta factor is initially defined on an absolutely convergent line by

$$\zeta_\alpha^*\Big(\tfrac{1}{2}+it\Big) = \frac{1}{2\pi i} \int_{(1)} \Gamma(u) Q(u) Y^u \zeta\Big(\tfrac{1}{2}+\alpha+u+it\Big)\, \mathrm{d}u. \tag{3.9}$$

There are two distinct contour operations. To produce a Dirichlet expansion, keep $\Re u = 1$, insert Lemma 3.1, and expand the three Dirichlet series there, with the AFE s-line initially at 2. For finite Dirichlet truncations move only the inverse-Mellin kernel to a fixed line

$$0 < c_u < \frac{1}{4}. \tag{3.10}$$

No pole is crossed. The uniform product bounds in Lemma 3.2 then remove the truncations. Thus the long zeta series is never expanded on the nonconvergent c_u-line.

For comparison with the original zeta function, instead keep the zeta factor in (3.9) unexpanded and move the collective u-line. The identity

$$\Gamma(u) R_M(u) = \frac{\Gamma(u+M+1)}{M!\, u} \tag{3.11}$$

shows that the poles $-1, \ldots, -M$ are killed. Its inverse Mellin weight is

$$W_M(x) = e^{-x} \sum_{j=0}^{M} \frac{x^j}{j!} = 1 + O_M(x^{M+1}) \qquad (x \longrightarrow 0). \tag{3.12}$$

The remaining polynomial factors in Q/R_M act by a fixed polynomial in $-x\frac{d}{dx}$ and have value one at $u=0$. Moving (3.9) to $\Re u = -M-1/2$ therefore crosses only $u=0$ and the moving zeta pole $u = 1/2-\alpha-it$. The latter is exponentially small by Stirling. Hence

$$\zeta_\alpha^*\Big(\tfrac{1}{2}+it\Big) = \zeta\Big(\tfrac{1}{2}+\alpha+it\Big) + O_{M,C}\Big((\log T)^{C_M}(T/Y)^{M+1/2}\Big) + O(e^{-cT}). \tag{3.13}$$

For fixed $B_0>2$, M can make this error smaller than any prescribed power.

Combining the two Mellin transforms, write

$$\mathscr{V}^\pm(\xi,x,t) = \frac{1}{(2\pi i)^2}\int_{(c_u)}\int_{(2)} \Gamma(u)\frac{\mathcal{G}(s,u)}{s} g_\pm(s,t)\xi^{-u}x^{-s}\,\mathrm{d}s\,\mathrm{d}u, \tag{3.14}$$

where $g_+ = g_{\beta,\gamma}$ and $g_- = g_{-\gamma,-\beta}$.

Lemma 3.2 (Full-contour double-weight bounds)**.** *For nonnegative integers i,j,r and every $A_0>0$,*

$$\xi^i x^j\left|\partial_\xi^i\partial_x^j\partial_t^r\mathscr{V}^\pm(\xi,x,t)\right| \ll_{i,j,r,A_0,M} (\log T)^{C_M}T^{-r}(1+\xi)^{-A_0}(1+x/T)^{-A_0}. \tag{3.15}$$

Every contour used in this bound is a complete infinite vertical line.

Proof. Stirling's formula gives

$$\partial_t^r g_\pm(s,t) \ll T^{\Re s - r}(1+|s|)^{O(r)}.$$

The factors e^{s^2} and $\Gamma(u)$ give Gaussian and exponential decay in the respective imaginary directions. Choose

$$0<\delta_u<1, \qquad 0<\delta_s<\tfrac{1}{2}-C/\log T.$$

For $\xi\ge 1$, move u arbitrarily far right. For $\xi\le 1$, move it to $\Re u=-\delta_u$; only $u=0$ is crossed. Similarly, for $x/T\ge 1$ move s right, while for $x/T\le 1$ move it to $\Re s = -\delta_s$; only $s=0$ is crossed. In the quadrant where both variables are small, retain both single residues and their common $(u,s)=(0,0)$ residue, which equals one. In either mixed quadrant, put the residue from the small-variable move on the right-shifted contour of the large variable. This gives

$$\mathscr{V}^\pm(\xi,x,t) \ll_{A_0} (1+\xi)^{-A_0}(1+x/T)^{-A_0}.$$

Normalized ξ- and x-derivatives insert polynomials in u and s; t-derivatives give the stated T^{-r}. Exponential decay justifies every move on full contours and avoids endpoint terms from hard height truncation. $\square$

4. Arithmetic decomposition and exact Poisson density

After inserting Lemma 3.1 into (3.9), the summands in the direct AFE piece, before the outer coefficient weight, are

$$l^{-1/2-\alpha}m^{-1/2-\beta}n^{-1/2-\gamma}\left(\frac{hn}{klm}\right)^{it}\mathscr{V}^+(l/Y,mn,t). \tag{4.1}$$

The dual piece is obtained by

$$(\beta,\gamma,\mathscr{V}^+)\mapsto(-\gamma,-\beta,\mathscr{V}^-)\quad\text{and multiplication by } X_{\beta,\gamma}(t). \tag{4.2}$$

The recovery estimate (3.13) permits us to work with these sums and return to the original cubic moment with an arbitrary power saving.

Put

$$hn-klm=r. \tag{4.3}$$

The terms with $r=0$ are the direct and dual diagonals. Suppose $r\neq 0$ and set

$$d=(h,km),\qquad h=dq,\qquad km=dv,\qquad r=d\rho. \tag{4.4}$$

Solutions exist only if $d\mid r$. Since $(q,v)=1$, the equation becomes

$$l\equiv-\rho\overline{v}\pmod q. \tag{4.5}$$

Thus every inverse is taken modulo a coprime integer, and Poisson summation in the fixed l-orientation has exact density

$$\frac{1}{q}=\frac{d}{h}. \tag{4.6}$$

We use this same orientation for both zero and nonzero frequencies.

More explicitly, if F is a smooth function of l, then

$$\sum_{l\equiv-\rho\overline{v}\ (q)}F(l)=\frac{1}{q}\sum_{a\in\mathbb{Z}}e\Big(-\frac{a\rho\overline{v}}{q}\Big)\int_{\mathbb{R}}F(y)e\Big(-\frac{ay}{q}\Big)\,\mathrm{d}y,\qquad e(x)=e^{2\pi ix}. \tag{4.7}$$

After the change of variables $x=kmy$, the combined Jacobian and progression density is

$$\frac{1}{qkm}=\frac{d}{hkm}. \tag{4.8}$$

For later use we record the exact zero mode, including the discrete endpoints. Choose $U\in C^\infty(\mathbb{R})$ with $U(x)=0$ for $x\le 1/2$ and $U(x)=1$ for $x\ge 1$. Multiplication by $U(l)U(n)$ does not change the discrete sum. With

$$y=kml,\qquad x=hn=y+r, \tag{4.9}$$

the direct zero mode for fixed h,k is

$$\begin{aligned}\mathcal{O}^+_{0,Y}(h,k)=\int_{\mathbb{R}}w(t)\sum_{m\ge1}\sum_{\substack{r\neq0\\ d\mid r}}d\,h^{-1/2+\gamma}k^{-1/2+\alpha}m^{-1+\alpha-\beta}\\ \times\int_{\max(0,-r)}^{\infty}y^{-1/2-\alpha-it}x^{-1/2-\gamma+it}\mathscr{V}^+\Big(\frac{y}{kmY},\frac{mx}{h},t\Big)\\ \times U\Big(\frac{y}{km}\Big)U\Big(\frac{x}{h}\Big)\,\mathrm{d}y\,\mathrm{d}t.\end{aligned} \tag{4.10}$$

The outside factor $a_h\overline{b_k}/\sqrt{hk}$ is not included in (4.10). Formula (4.10) follows directly from (4.6) after changing $\mathrm{d}l$ to $\mathrm{d}y/(km)$; in particular, the factor d has not been suppressed. The dual zero mode is obtained by (4.2).

Let $\widetilde{\mathcal{O}}^{+}_{0,Y}$ be (4.10) with both U factors deleted, initially wherever the resulting integrals and sums converge absolutely and subsequently by the continuation constructed in Section 6. Define $\widetilde{\mathcal{O}}^{-}_{0,Y}$ in the same way.

For the nonzero-frequency analysis, decompose all five arithmetic variables into dyadic blocks

$$h \asymp H_0, \quad k \asymp K_0, \quad l \asymp L_0, \quad m \asymp M_0, \quad n \asymp N_0$$

and put

$$S = H_0N_0 + K_0L_0M_0. \tag{4.11}$$

Repeated integration by parts in t shows that a nonnegligible block must satisfy

$$H_0N_0 \asymp K_0L_0M_0, \qquad |r| \ll ST^{-1+\varepsilon}. \tag{4.12}$$

Consequently

$$S \asymp H_0N_0 \asymp K_0L_0M_0 \asymp (H_0K_0L_0M_0N_0)^{1/2}. \tag{4.13}$$

The AFE conductor also gives

$$M_0N_0 \ll T^{1+\varepsilon}, \tag{4.14}$$

while the long smoothing has $L_0 \ll YT^{\varepsilon}$. The logarithmic number of blocks is absorbed into T^{ε}.

5. The nonzero Poisson frequencies

We prove the estimate that determines the range of Theorem 2.1. The argument uses only finite completion and the classical Weil bound [8]; it does not correlate the two coefficient sequences.

Throughout this section the convention following (2.1) is in force. Thus the constants behind T^{ε} may depend on the fixed C, B_0, M, on finitely many seminorms (2.2)–(2.3), and on the fixed completion order, but not on the dyadic block parameters or on H, K, T. The parameter η enters only when (2.8) is invoked in the global summation. The auxiliary ε_0 used below is chosen only after the completion order is known and is then absorbed into the final ε; it is not an additional uniformity parameter.

For a dyadic block as in Section 4, the t-integral and the change of variables in (4.8) must be kept together. Along $hn - klm = r$, the phase is $x/(x+r)$, and

$$x\frac{d}{dx}\left(t\log\frac{x}{x+r}\right) \ll \frac{T|r|}{S} \ll T^{\varepsilon}. \tag{5.1}$$

Consequently the Poisson integral has size TS, with normalized derivatives bounded by $TS\,T^{\varepsilon}$. The factor T here is essential.

For $d = (h, km)$ as in (4.4), integration by parts in the Poisson integral restricts a nonzero Fourier frequency to

$$0 < |a| \le A_d, \qquad A_d \ll \frac{H_0K_0M_0}{dS}T^{\varepsilon} \asymp \frac{S}{dL_0N_0}T^{\varepsilon}. \tag{5.2}$$

Also

$$0 < |\rho| \ll \frac{R_0}{d}, \qquad R_0 = ST^{-1+\varepsilon}. \tag{5.3}$$

If either upper bound is less than one, the corresponding block is empty.

Lemma 5.1 (Incomplete reciprocal sum)**.** *Let $q, c \geq 1$, and let F be supported in an interval of length X with $F^{(j)} \ll_j ZX^{-j}$. Then*

$$\sum_{(u,q)=1} F(u)e\left(\frac{c\overline{u}}{q}\right) \ll_\varepsilon Zq^\varepsilon \left\{Xq^{-1/2}(c,q)^{1/2} + q^{1/2}(c,q)^{1/2}\right\}. \tag{5.4}$$

Moreover,

$$\sum_{q \asymp Q} (c,q)^{1/2} \ll_\varepsilon Qc^\varepsilon. \tag{5.5}$$

Proof. With the Fourier-transform convention compatible with $e(x) = e^{2\pi i x}$, finite completion gives

$$\sum_{(u,q)=1} F(u)e\left(\frac{c\overline{u}}{q}\right) = \frac{1}{q}\sum_{n\in\mathbb{Z}} S(c,n;q)\widehat{F}(n/q),$$

where

$$S(c,n;q) = \sideset{}{^*}\sum_{u \bmod q} e\left(\frac{c\overline{u} + nu}{q}\right).$$

For every fixed $A \geq 2$, repeated integration by parts gives

$$|\widehat{F}(\xi)| \ll_A ZX(1 + X|\xi|)^{-A}, \qquad \sum_{n\in\mathbb{Z}} |\widehat{F}(n/q)| \ll_A Z(X+q).$$

The Weil bound $|S(c,n;q)| \leq \tau(q)(c,n,q)^{1/2}q^{1/2}$, together with $(c,n,q)^{1/2} \leq (c,q)^{1/2}$ for every n, therefore proves (5.4), uniformly for both $X \leq q$ and $X > q$. Thus the factor $(c,q)^{1/2}$ occurs in both the $Xq^{-1/2}$ term and the $q^{1/2}$ term. The term $n = 0$ is the Ramanujan sum and obeys the same majorant. Finally, $(c,q)^{1/2} \leq \sum_{e|(c,q)} e^{1/2}$ gives

$$\sum_{q\asymp Q} (c,q)^{1/2} \leq \sum_{e|c} e^{1/2}\#\{q \asymp Q : e \mid q\} \ll Q\sum_{e|c} e^{-1/2} \ll_\varepsilon Qc^\varepsilon,$$

which is (5.5). □

For the coefficient block put

$$\mathcal{A}_{H_0} = \max_{h\asymp H_0} |a_h|, \qquad \mathcal{B}_{K_0} = \sum_{k\asymp K_0} |b_k|. \tag{5.6}$$

Proposition 5.2 (A fixed-orientation block estimate)**.** *The contribution of the nonzero frequencies obtained by Poisson summation in l satisfies*

$$E^{(l)}(H_0, K_0, L_0, M_0, N_0) \ll_\varepsilon T^\varepsilon \mathcal{A}_{H_0}\mathcal{B}_{K_0}H_0^{1/2}(H_0 + M_0). \tag{5.7}$$

Proof. Choose the dyadic partition using fixed functions

$$\omega_H, \omega_K, \omega_L, \omega_M, \omega_N \in C_c^\infty((1/2, 3)).$$

All their derivatives are bounded independently of the block parameters. We first expose the smooth weight to which Lemma 5.1 is applied.

Fix d, a, ρ and k, and write

$$g = (d, k), \qquad d = g\delta, \qquad k = gk_1, \qquad m = \delta m_1. \tag{5.8}$$

The equality $(d, k) = g$ implies $(\delta, k_1) = 1$: for every prime p, the minimum of the p-adic valuations of d/g and k/g is zero. Since $d \mid km$, it follows that $\delta \mid m$, so m_1 is an integer. If $h = dq$, then

$$(h, km) = d \quad \Longleftrightarrow \quad (q, k_1 m_1) = 1, \qquad \frac{km}{d} = k_1 m_1. \tag{5.9}$$

All substitutions are uniquely determined by d, h, k, m; conversely the right side implies the original gcd condition. Thus this is a bijective parametrization, with neither omission nor repeated counting.

Put $r = d\rho$ and, after the change of variables $x = kml$ from (4.8), set

$$\ell_x = \frac{x}{km}, \qquad n_x = \frac{x + d\rho}{h}. \tag{5.10}$$

For the direct AFE piece define the exact normalized amplitude

$$\begin{aligned}
\mathscr{W}^+_{h,k,m,d\rho}(x,t) = {} & \frac{(H_0K_0L_0M_0N_0)^{1/2} H_0K_0M_0}{hkm\,(hk\ell_x m n_x)^{1/2}} \\
& \times \omega_H(h/H_0)\omega_K(k/K_0)\omega_L(\ell_x/L_0) \\
& \times \omega_M(m/M_0)\omega_N(n_x/N_0)\ell_x^{-\alpha} m^{-\beta} n_x^{-\gamma} w(t) \left(\frac{x + d\rho}{x}\right)^{it} \\
& \times \mathscr{V}^+\left(\frac{x}{kmY}, \frac{m(x + d\rho)}{h}, t\right).
\end{aligned} \tag{5.11}$$

The cutoffs make the integrand zero unless $x > 0$ and $x + d\rho > 0$. The dual amplitude is obtained by the exact substitution (4.2); its additional factor $X_{\beta,\gamma}(t)$ is independent of m_1 and is $T^{O(1/\log T)}$ on the support of w.

For either AFE sign let

$$\mathfrak{F}^\pm_{d,a,\rho,q,k}(m_1) = \frac{1}{TS}\int_{\mathbb{R}}\int_{\mathbb{R}} \mathscr{W}^\pm_{dq,k,\delta m_1,d\rho}(x,t)\, e\Big(-\frac{ax}{qk\delta m_1}\Big)\,\mathrm{d}x\,\mathrm{d}t. \tag{5.12}$$

Combining (4.1), (4.7), and (4.8), the contribution for fixed d, a, ρ, q, k, m_1, before the reciprocal phase is summed, is exactly

$$\frac{dTS}{(H_0K_0L_0M_0N_0)^{1/2} H_0K_0M_0} a_{dq}\overline{b_k}\, \mathfrak{F}^\pm_{d,a,\rho,q,k}(m_1)\, e\left(-\frac{a\rho\overline{k_1 m_1}}{q}\right), \tag{5.13}$$

up to the common sign convention for the two exponentials. The factor d is precisely the numerator of the density $d/(hkm)$ and has not been hidden in the smooth weight.

We now verify the completion hypotheses. Fix an auxiliary $0 < \varepsilon_0 < \varepsilon$ and invoke the localization, frequency truncation, and divisor estimates in this proof with ε_0 in place of ε; its size will be fixed after the finite completion order is known. The support is contained in a fixed multiple of an interval of length

$$X_m = \frac{M_0}{\delta} = \frac{M_0 g}{d}. \tag{5.14}$$

If this interval contains an integer, then $X_m \gg 1$; otherwise the sum is empty. On the x-support one has $x \asymp K_0 L_0 M_0 \asymp S$, where $S = H_0 N_0 + K_0 L_0 M_0$ is fixed throughout the block. Put $D_x = x\partial_x$. Since $|d\rho|/x \ll T^{-1+\varepsilon_0}$, the identity

$$\log \frac{x}{x + d\rho} = -\int_0^{d\rho} \frac{\mathrm{d}v}{x+v}$$

gives, for every fixed $j \geq 0$,

$$\left| D_x^j \left[t \log \frac{x}{x+d\rho} \right] \right| \ll_j \frac{T|d\rho|}{S} \ll T^{\varepsilon_0}. \tag{5.15}$$

The segment $x+v$ remains comparable with S, also when $d\rho < 0$. Repeated differentiation of the exponential therefore costs only $T^{O_j(\varepsilon_0)}$.

At fixed x the t-oscillation is independent of m_1, while

$$m_1\partial_{m_1}\ell_x = -\ell_x, \qquad m_1\partial_{m_1} n_x = 0, \qquad m_1\partial_{m_1}\left[t\log\frac{x}{x+d\rho}\right] = 0. \tag{5.16}$$

If the moving support is instead fixed by $x = k\delta m_1 L_0 y$, then

$$m_1\partial_{m_1}|_y = m_1\partial_{m_1}|_x + D_x,$$

so (5.15) gives the same conclusion. In particular, this change does not differentiate the fixed block-scale S. For

$$\xi = \frac{x}{kmY}, \qquad \chi = \frac{m(x+d\rho)}{h},$$

one has

$$m_1\partial_{m_1}\xi = -\xi, \qquad m_1\partial_{m_1}\chi = \chi. \tag{5.17}$$

Thus (3.15) controls every derivative falling on $\mathscr{V}^\pm$. Derivatives of the dyadic cutoffs cost X_m^{-1}, and all algebraic powers have bounded logarithmic derivatives on their supports. Finally,

$$\left|\frac{ax}{qkm}\right| \ll \frac{|a|dS}{H_0K_0M_0} \ll T^{\varepsilon_0} \tag{5.18}$$

by (5.2); hence every fixed number of derivatives of the inner Poisson Fourier phase is also $T^{O_j(\varepsilon_0)}$.

The x-support has length $O(S)$ and the t-support length $O(T)$. Differentiating under the integrals in (5.12) therefore yields, for every fixed $j \geq 0$,

$$\left|(m_1\partial_{m_1})^j \mathfrak{F}^\pm_{d,a,\rho,q,k}(m_1)\right| \ll_j T^{O_j(\varepsilon_0)}, \qquad \left|\frac{d^j}{dm_1^j}\mathfrak{F}^\pm_{d,a,\rho,q,k}(m_1)\right| \ll_j T^\varepsilon X_m^{-j}. \tag{5.19}$$

Here the localization loss ε_0 is chosen sufficiently small in terms of the final ε and the fixed number of derivatives used in completion. No derivative order depends on T.

Since $(q, k_1) = 1$, the coefficient in the reciprocal phase may be taken as $c = \pm a\rho\overline{k_1} \pmod q$, for which $(c, q) = (a\rho, q)$. Applying Lemma 5.1 to the actual weight gives

$$\sum_{\substack{m_1 \asymp M_0 g/d \\ (m_1,q)=1}} \mathfrak{F}^{\pm}_{d,a,\rho,q,k}(m_1) e\left(\frac{\pm a\rho\overline{k_1 m_1}}{q}\right) \ll T^{\varepsilon}\left\{\frac{M_0 g}{d} q^{-1/2}(a\rho, q)^{1/2} + q^{1/2}(a\rho, q)^{1/2}\right\}. \tag{5.20}$$

Put $Q_d = H_0/d$. By (5.5), restriction to $(q, k_1) = 1$ can only decrease the majorant. Applying (5.5) separately to the two terms in (5.20) gives

$$\frac{M_0 g}{d} Q_d^{-1/2} \sum_{q \asymp Q_d} (a\rho, q)^{1/2} \ll_{\varepsilon} \frac{M_0 g}{d} Q_d^{1/2} (a\rho)^{\varepsilon}$$

and

$$Q_d^{1/2} \sum_{q \asymp Q_d} (a\rho, q)^{1/2} \ll_{\varepsilon} Q_d^{3/2} (a\rho)^{\varepsilon}.$$

Consequently,

$$\sum_{q \asymp Q_d} \left(\frac{M_0 g}{d} q^{-1/2} + q^{1/2}\right)(a\rho, q)^{1/2} \ll T^{\varepsilon}\left(\frac{M_0 g H_0^{1/2}}{d^{3/2}} + \frac{H_0^{3/2}}{d^{3/2}}\right). \tag{5.21}$$

Indeed these are respectively $X_m Q_d^{1/2}$ and $Q_d^{3/2}$. The factor $(a\rho)^{\varepsilon}$ is absorbable because all retained dyadic parameters are bounded by a fixed power of T; the auxiliary exponent is chosen before relabelling the total loss as T^{ε}.

Take the supremum norm of a_{dq} and sum over k. The sets $\{k : (d, k) = g\}$ partition the k-block. The second term of (5.21) is independent of g, and in the first we use only $g \le d$. Therefore

$$\sum_{\substack{h \asymp H_0,\, k \asymp K_0,\, m \asymp M_0 \\ (h,km)=d}} a_h \overline{b_k}\, \mathfrak{F}^{\pm}_{d,a,\rho,h/d,k}(m/\delta) e\left(\frac{\pm a\rho\overline{km/d}}{h/d}\right) \ll T^{\varepsilon} \mathcal{A}_{H_0} \mathcal{B}_{K_0}\left(\frac{H_0^{3/2}}{d^{3/2}} + \frac{M_0 H_0^{1/2}}{d^{1/2}}\right). \tag{5.22}$$

We finish by multiplying every scalar factor. After the density numerator d is removed from (5.13), the original Dirichlet normalization, the remaining progression density and Jacobian, and the actual t, x-integral size give

$$\mathcal{C}_0 = \frac{TS}{(H_0K_0L_0M_0N_0)^{1/2}H_0K_0M_0}. \tag{5.23}$$

More explicitly, the original normalization is $(H_0K_0L_0M_0N_0)^{-1/2}$, the progression density is d/H_0, the $x = kml$ Jacobian costs $(K_0M_0)^{-1}$, and the Poisson integral has size TS. For fixed d, the density numerator, the number of frequencies, and the number of nonzero shifts contribute

$$d \cdot \frac{A_0}{d} \cdot \frac{R_0}{d} = \frac{A_0R_0}{d}, \qquad A_0 = \frac{H_0K_0M_0}{S}T^{\varepsilon_0}, \quad R_0 = \frac{S}{T}T^{\varepsilon_0}. \tag{5.24}$$

Multiplying (5.22)–(5.24) and extending the d-sum to all positive integers gives

$$\begin{aligned} E^{(l)}(H_0, K_0, L_0, M_0, N_0) &\ll T^{\varepsilon}\mathcal{C}_0A_0R_0\mathcal{A}_{H_0}\mathcal{B}_{K_0} \\ &\qquad \times \left(H_0^{3/2}\sum_{d\geq 1} d^{-5/2} + M_0H_0^{1/2}\sum_{d\geq 1} d^{-3/2} \right). \end{aligned} \tag{5.25}$$

Both d-series converge. Finally, by (4.13),

$$\mathcal{C}_0A_0R_0 = \frac{S}{(H_0K_0L_0M_0N_0)^{1/2}}T^{O(\varepsilon_0)} \asymp T^{O(\varepsilon_0)}. \tag{5.26}$$

Choosing ε_0 in terms of the final ε proves (5.7). $\square$

Proposition 5.3 (Nonzero-frequency estimate)**.** *With the fixed l-Poisson orientation,*

$$\mathcal{R}^{(l)}_{\neq 0} \ll_{\varepsilon} T^{\varepsilon}\|a\|_{\infty}H^{3/2}\sum_{k\leq K}|b_k|. \tag{5.27}$$

In particular, under (2.9),

$$\mathcal{R}^{(l)}_{\neq 0} \ll_{\varepsilon} T^{\varepsilon}KH^{3/2}. \tag{5.28}$$

Proof. For a nonempty nonzero-frequency block, $r = d\rho \neq 0$ and (4.12) imply

$$S \gg dT^{1-\varepsilon}. \tag{5.29}$$

Since $N_0 \asymp S/H_0$ and $M_0N_0 \ll T^{1+\varepsilon}$,

$$M_0 \ll \frac{H_0T^{1+\varepsilon}}{S} \ll \frac{H_0}{d}T^{2\varepsilon} \leq H_0T^{2\varepsilon}. \tag{5.30}$$

Substitution in (5.7), followed by the logarithmic dyadic sums and the two AFE pieces, gives (5.27). The divisor bounds then give (5.28). $\square$

Remark 5.4. Poisson summation in another variable gives a separately valid estimate for a different decomposition, but its zero frequency is different. We do not take a minimum over Poisson orientations. The fixed orientation used in Proposition 5.3 is exactly the one whose zero mode is evaluated in the following sections.

6. Endpoint deletion and the full-kernel bridge

The zero mode (4.10) is initially a continuous integral with the discrete restrictions $l, n \geq 1$ retained by U. We first delete the artificial regions $0 < l < 1$ and $0 < n < 1$ on a genuine convergence patch. We then construct the exact continuation needed to reach all small shifts.

6.1. Fourier decay and the endpoint lemma.

Lemma 6.1 (Fourier–conductor bound)**.** *Uniformly for $x, y > 0$, $x \neq y$, and every $A_0, J > 0$,*

$$\left|\int_{\mathbb{R}} w(t)(x/y)^{it}\mathscr{V}^{\pm}\left(\frac{y}{kmY}, \frac{mx}{h}, t\right)\,\mathrm{d}t\right| \ll_{A_0,J} T(\log T)^{C_M}\left(1+\frac{y}{kmY}\right)^{-A_0} \times\left(1+\frac{mx}{hT}\right)^{-A_0}(1+\Delta|\log(x/y)|)^{-J}. \tag{6.1}$$

Proof. Integrate by parts J times in t and use (2.1) and (3.15). Every derivative costs at most Δ^{-1}, and the support of w has length $O(T)$. □

Proposition 6.2 (Endpoint deletion on the good half-space)**.** *Assume* (2.8)*. For every $D > 0$, after summing over $h \leq H$, $k \leq K$ with the outside coefficient weights, one has*

$$\mathcal{O}^{\pm}_{0,Y} - \widetilde{\mathcal{O}}^{\pm}_{0,Y} = O_D(T^{-D}) \tag{6.2}$$

when

$$\Re(\beta-\gamma) \geq 0, \qquad |\Re\alpha| + |\Re\beta| + |\Re\gamma| \leq C/\log T. \tag{6.3}$$

The source expressions retain the complete contours in (3.14)*.*

Proof. It suffices to treat

$$\mathcal{E}_l = \{y < km\}, \qquad \mathcal{E}_n = \{x < h\}. \tag{6.4}$$

Put $\delta_0 = \eta/4$ and $\varepsilon_0 = \eta/8$. On $\mathcal{E}_l$, suppose simultaneously that

$$|\log(x/y)| \leq T^{-1+\delta_0}, \qquad \frac{mx}{hT} \leq T^{\varepsilon_0}. \tag{6.5}$$

Then $x \asymp y$, and, with $r = d\rho$, $h = dq$, $km = dv$,

$$y \gg d|\rho|T^{1-\delta_0}, \qquad v \gg |\rho|T^{1-\delta_0}, \qquad m|\rho| \ll qT^{\delta_0+\varepsilon_0}. \tag{6.6}$$

Since $v \leq km$ and $q \leq h \leq H$, this forces

$$HK \gg T^{1-2\delta_0-\varepsilon_0} = T^{1-5\eta/8}. \tag{6.7}$$

But (2.8) and $H \geq 1$ give $HK \leq T^{1-\eta}$, a contradiction for large T. Every point of $\mathcal{E}_l$ therefore has either Fourier saving $|\log(x/y)| > T^{-1+\delta_0}$ or conductor saving $mx/(hT) > T^{\varepsilon_0}$. The orders A_0, J in (6.1) absorb every remaining polynomial factor.

On $\mathcal{E}_n$, $0 < x < h$ and $r = d\rho \neq 0$ imply

$$|\log(x/y)| \gg \min(1, |r|/h) \gg 1/H. \tag{6.8}$$

The main range gives $H \leq T^{2(1-\eta)/3}$, and hence

$$\Delta|\log(x/y)| \gg T^{(1+2\eta)/3}/\log T. \tag{6.9}$$

Repeated integration by parts again gives an arbitrary power saving, once the endpoint mass is shown to be only polynomial.

We record the singular-layer summation because it is where the gcd density is needed. For fixed m and a dyadic l-scale, the values

$$n = \frac{kml + d\rho}{h}$$

form a mesh of spacing d/h, exactly matching (4.6). A boundary layer meeting $n = 0$ has $kml \asymp d|\rho|$. On writing $n = (d/h)x$ and $x = |\rho|z$, the square-root Jacobian is

$$|\rho|^{-1/2} \int_0^{O(|\rho|)} x^{-1/2}(\cdots)\,\mathrm{d}x = \int_0^{O(1)} z^{-1/2}(\cdots)\,\mathrm{d}z. \tag{6.10}$$

Thus no unrecorded sum over ρ and no lost $|\rho|^{-1/2}$ occurs.

For the potentially worst case $r < 0$, $x < h$, partial summation in $r = d\rho$ and then in m gives, up to T^{ε},

$$\sum_{\rho \geq 1} y^{-1/2-\Re\alpha} \left(1 + \frac{y}{kmY}\right)^{-A_0} \ll d^{-1}(kmY)^{1/2-\Re\alpha}, \tag{6.11}$$

$$\sum_{m \geq 1} m^{-1/2-\Re\beta} \left(1 + \frac{mx}{hT}\right)^{-A_0} \ll (hT/x)^{1/2-\Re\beta}. \tag{6.12}$$

After multiplication by $x^{-1/2-\Re\gamma}$, the lower-end behavior is

$$x^{-1+\Re(\beta-\gamma)}. \tag{6.13}$$

It is integrable in (6.3); on the boundary, the last factor in (6.1) makes the logarithmic endpoint integrable for $J > 1$. All remaining sums have fixed polynomial size and are absorbed by (6.9). This proves (6.2). $\square$

The ordinary dominated-convergence proof stops at $\Re(\beta - \gamma) = 0$: its far dyadic majorant has residual size

$$M_1^{\Re(\gamma-\beta)}\{1 + \log(M_1/T)\}^{-J} \qquad (M_1 \gg T). \tag{6.14}$$

We therefore continue only the complete assembled error, rather than a single endpoint-deleted branch.

6.2. **Source normality with the full kernel.** In this subsection use external shifts A, B, C_1 in place of α, β, γ in the kernel. The continuation requires only one nonempty patch; for definiteness take $\kappa_0 = 1$, $c_1 = 5$, and put

$$\Omega_g = \left\{ \Re B > \frac{\kappa_0}{\log T}, \quad \Re C_1 < -\frac{\kappa_0}{\log T}, \quad |A| + |B| + |C_1| < \frac{c_1}{\log T} \right\}. \tag{6.15}$$

Fix B, C_1 there and let A vary in the connected strip

$$\Re A < \frac{1}{2} - \frac{c_u}{2}. \tag{6.16}$$

Lemma 6.3 (Full-kernel source normality)**.** *For each of the four pairs*

$$(\textit{direct or dual AFE sign},\ \operatorname{sgn}\rho),$$

the endpoint-deleted expressions, with the finite regulators introduced below, converge locally uniformly in the strip (6.16)*, uniformly for* (B, C_1) *on compact subsets of* (6.15)*. Their limits are holomorphic in* A*, as are any fixed number of* A*-derivatives.*

Proof. Keep the complete kernel $\mathscr{V}(l/Y, mn, t)$ and do not freeze its s variable. For the exact-cutoff source, $l, n \geq 1/2$. If $F(n)$ is the nonnegative product-tail majorant from Lemma 3.2, then on either half-mesh of spacing $\delta = d/h$,

$$\delta \sum_{\rho : n \geq 1/2} F(n) \ll \int_{1/2}^{\infty} F(v)\,\mathrm{d}v + \delta \sup_{v \geq 1/2} F(v). \tag{6.17}$$

This gives, before the finite outer sums, a fixed logarithmic power times

$$T \int_{1/2}^{\infty} l^{-1/2-a_0} (1 + l/Y)^{-P}\,\mathrm{d}l \sum_{m \geq 1} m^{-1/2-b_0} \times \left\{ \int_{1/2}^{\infty} n^{-1/2-c_0} (1 + mn/T)^{-Q}\,\mathrm{d}n + (1 + m/T)^{-Q} \right\}. \tag{6.18}$$

where $a_0 = \Re A$, $b_0 = \Re B$, and $c_0 = \Re C_1$. On compact subsets of (6.16), this is polynomial in T and locally uniform; the residual m-sum is $O(\log T)$ on the good patch. The same is true after the dual substitution $(B, C_1) \mapsto (-C_1, -B)$.

For completeness, consider the worst $n = 0$ discrepancy with $r = -d\rho$, $z = x/(d\rho) \leq 1$, and restore the outside coefficient weight. Its full algebraic prefactor is

$$|a_h b_k| d^{1-a_0-c_0} h^{-1+c_0} k^{-1+a_0} m^{-1+a_0-b_0} \rho^{-a_0-c_0} \times \int_0^1 z^{-1/2-c_0} \left(1 + \frac{md\rho z}{hT}\right)^{-Q} \left(1 + \frac{d\rho}{kmY}\right)^{-P} (1 + |\log z|)^{-J}\,\mathrm{d}z. \tag{6.19}$$

For $X = md\rho/(hT)$,

$$\int_0^1 z^{-1/2-c_0} (1+Xz)^{-Q} (1+|\log z|)^{-J}\,\mathrm{d}z \ll (1+X)^{-1/2+c_0} \{1+\log(2+X)\}^{-J}. \tag{6.20}$$

In the range $X > 1$, summing m and ρ gives at most

$$\left(\frac{hT}{d}\right)^{1/2-c_0}\left(\frac{kY}{d}\right)^{1/2-a_0}\sum_{m\geq 1} m^{-1-b_0+c_0}. \tag{6.21}$$

The last sum is $O_{\kappa_0}(\log T)$. Multiplication by the prefactor in (6.19) cancels every power of d and leaves

$$|a_h b_k|(hk)^{-1/2}T^{1/2-c_0}Y^{1/2-a_0}\sum_{m\geq 1} m^{-1-b_0+c_0}. \tag{6.22}$$

After the h, k sums and the Fourier prefactor, the $X > 1$ discrepancy is

$$\ll T\Delta^{-J}T^{1/2-c_0}Y^{1/2-a_0}(HK)^{1/2}T^{\varepsilon}\log T, \tag{6.23}$$

and is $O_N(T^{-N})$ after increasing J. This exact cancellation is uniform over all gcd blocks. The range $X \leq 1$ has $m\rho \leq hT/d$ and therefore only polynomial mass on compact external A-strips. For $z > 1$, $x < h = dq$ gives

$$\left|\log\frac{x}{x+d\rho}\right| = \log(1+z^{-1}) \gg q^{-1} \geq H^{-1}. \tag{6.24}$$

In both cases an arbitrarily high power in the Fourier bound absorbs the polynomial mass. On far m-scales, the residual exponent is

$$a_0 - \Re(B-C_1) - \frac{1}{2} \leq -\frac{c_u}{2} - \Re(B-C_1), \tag{6.25}$$

so the dyadic sum is uniform on the good patch. The estimates remain valid after A-derivatives, which only insert logarithms. This proves local normality and holomorphy. $\square$

6.3. **Finite regulators and genuine Tonelli domains.**

Proposition 6.4 (Four-branch regulator–Tonelli bridge)**.** *Fix (B, C_1) in a compact subset of* (6.15)*, keep $\Re u = c_u$, and put $\lambda = A + u$. Define*

$$z_{\mathrm{d}} = \lambda + C_1 + s, \qquad \nu_{\mathrm{d}} = 1 - \lambda + B + s, \tag{6.26}$$

$$z_{\mathrm{q}} = \lambda - B + s, \qquad \nu_{\mathrm{q}} = 1 - \lambda - C_1 + s. \tag{6.27}$$

Here d *denotes the direct AFE piece and* q *the dual piece. In each of the four branches impose*

$$\begin{aligned} m, |\rho| \leq R, &\qquad R^{-1} \leq l, n \leq R, \\ e^{-\kappa(m+|\rho|+l+n)}, &\qquad R \geq 2, \quad 0 < \kappa \leq 1. \end{aligned} \tag{6.28}$$

Then the source limits obtained in the order

$$V \longrightarrow \infty, \qquad R \longrightarrow \infty, \qquad \kappa \longrightarrow 0^{+} \tag{6.29}$$

exist locally uniformly in the connected strip (6.16) *and are holomorphic in the original shift A. The corresponding destination limits are absolutely convergent in the following four nonempty domains. The two negative branches use the common corridor*

$$\frac{1}{2} < \Re\lambda < \frac{1}{2} + \Re(B - C_1). \tag{6.30}$$

For the direct positive and negative branches, respectively, the remaining conditions are

$$\begin{aligned} &\Re\lambda < \frac{1}{2}, \\ &\Re s > \max\{1-\Re(\lambda+C_1), \Re(\lambda-B)\}. \end{aligned} \tag{6.31}$$

$$\max\{1-\Re(\lambda+C_1), \Re(\lambda-B)\} < \Re s < \frac{1}{2} - \Re C_1. \tag{6.32}$$

For the dual positive and negative branches, respectively, they are

$$\begin{aligned} &\Re\lambda < \frac{1}{2}, \\ &\Re s > \max\{1-\Re(\lambda-B), \Re(\lambda+C_1)\}. \end{aligned} \tag{6.33}$$

$$\max\{1-\Re(\lambda-B), \Re(\lambda+C_1)\} < \Re s < \frac{1}{2} + \Re B. \tag{6.34}$$

For finite R, κ, V, the source and destination are the explicit objects defined in the proof and satisfy an exact rectangle identity, including its two horizontal s-edges. After $V \to \infty$ those edges vanish and the complete regulated source and destination agree exactly. After the remaining limits in (6.29)*, this equality extends from the relevant Tonelli open set to* (6.16) *by the one-variable identity theorem in A, with B, C_1 fixed. No continuation in the separate symbols $A+u$ and s is used.*

Proof. We first define the two sides before taking any limit. For $\epsilon \in \{\mathrm{d}, \mathrm{q}\}$ put

$$\begin{array}{c|ccc} \epsilon & B_\epsilon & C_\epsilon & \mathcal{X}_\epsilon(t) \\ \hline \mathrm{d} & B & C_1 & 1 \\ \mathrm{q} & -C_1 & -B & X_{B,C_1}(t). \end{array} \tag{6.35}$$

Thus $z_\epsilon = \lambda + C_\epsilon + s$ and $\nu_\epsilon = 1 - \lambda + B_\epsilon + s$. For $\rho \geq 1$ and $\mathfrak{r} = d\rho$, where $d = (h, km)$, define the physical variables

$$\begin{array}{c|cc} \sigma & y_\sigma(v) & x_\sigma(v) \\ \hline + & v & v+\mathfrak{r} \\ - & v+\mathfrak{r} & v \end{array} \qquad (v>0), \qquad l_\sigma(v) = \frac{y_\sigma(v)}{km}, \quad n_\sigma(v) = \frac{x_\sigma(v)}{h}. \tag{6.36}$$

Let $[a]_V$ denote the upward segment $a - iV$ to $a + iV$, and set

$$E_{R,\kappa,\sigma}(m,\rho,v) = \mathbf{1}_{\{m,\rho\leq R\}} \mathbf{1}_{\{R^{-1}\leq l_\sigma(v), n_\sigma(v)\leq R\}} e^{-\kappa\{m+\rho+l_\sigma(v)+n_\sigma(v)\}}. \tag{6.37}$$

For $\lambda = A + u$, define

$$\begin{aligned} \mathfrak{J}^{R,\kappa}_{\epsilon,\sigma}(s,u;t,m,\rho) := d\, h^{-1/2+C_\epsilon+s} k^{-1/2+\lambda} Y^u \\ \times m^{-1+\lambda-B_\epsilon-s} \int_0^\infty E_{R,\kappa,\sigma}(m,\rho,v) \\ \times y_\sigma(v)^{-1/2-\lambda-it} x_\sigma(v)^{-1/2-C_\epsilon+it-s}\,\mathrm{d}v. \end{aligned} \tag{6.38}$$

For fixed h, k, the finite source is

$$\mathscr{S}^{R,\kappa,V}_{\epsilon,\sigma}(A;h,k) := \int_{\mathbb{R}} w(t) \sum_{m\leq R} \sum_{\rho\leq R} \frac{1}{(2\pi i)^2} \int_{[c_u]_V} \int_{[2]_V} \Gamma(u) \frac{\mathcal{G}(s,u)}{s}$$

$$\times \mathcal{X}_\epsilon(t) g_{B_\epsilon,C_\epsilon}(s,t)\mathfrak{J}^{R,\kappa}_{\epsilon,\sigma}(s,u;t,m,\rho)\,\mathrm{d}s\,\mathrm{d}u\,\mathrm{d}t. \tag{6.39}$$

This is exactly the σ-branch of the endpoint-deleted zero mode after inserting the truncated coupled kernel (3.14); all sums and physical integrals are finite, so their order is immaterial. On a compact subset of the relevant Tonelli open set choose a single line $\Re s = \sigma_{\epsilon,\sigma} > 0$ satisfying the corresponding strict inequalities in (6.31)–(6.34). The finite destination is the same displayed expression with $[2]_V$ replaced by $[\sigma_{\epsilon,\sigma}]_V$:

$$\mathscr{T}^{R,\kappa,V}_{\epsilon,\sigma}(A;h,k) := \mathscr{S}^{R,\kappa,V}_{\epsilon,\sigma}(A;h,k)\big|_{[2]_V \mapsto [\sigma_{\epsilon,\sigma}]_V}. \tag{6.40}$$

No beta quotient or infinite arithmetic series is part of (6.40).

Write $\mathfrak{r} = d|\rho| > 0$. After the endpoint cutoffs have been deleted, the following four unregulated continuous integrals determine the Tonelli conditions. They are displayed separately because their zero endpoints differ; their beta evaluation is used only later, after all variable regulators have been removed.

For the direct, positive-ρ branch, $x = y + \mathfrak{r}$ and

$$\begin{aligned} I_{\mathrm{d},+}(\mathfrak{r}) &:= \int_0^\infty y^{-1/2-\lambda-it}(y+\mathfrak{r})^{-1/2-C_1+it-s}\,\mathrm{d}y \\ &= \mathfrak{r}^{-z_{\mathrm{d}}}\frac{\Gamma(\frac12-\lambda-it)\Gamma(z_{\mathrm{d}})}{\Gamma(\frac12+C_1-it+s)}. \end{aligned} \tag{6.41}$$

At $y = 0$ this requires $\Re\lambda < 1/2$; at infinity it first requires $\Re z_{\mathrm{d}} > 0$. The subsequent ρ-sum strengthens the latter to $\Re z_{\mathrm{d}} > 1$. On that half-plane the gcd exponent $1 - z_{\mathrm{d}}$ has negative real part, and the remaining m-sum is absolutely convergent when $\Re\nu_{\mathrm{d}} > 1$. These conditions are exactly (6.31).

For the direct, negative-ρ branch, put $y = x + \mathfrak{r}$. Then

$$\begin{aligned} I_{\mathrm{d},-}(\mathfrak{r}) &:= \int_0^\infty (x+\mathfrak{r})^{-1/2-\lambda-it}x^{-1/2-C_1+it-s}\,\mathrm{d}x \\ &= \mathfrak{r}^{-z_{\mathrm{d}}}\frac{\Gamma(\frac12-C_1+it-s)\Gamma(z_{\mathrm{d}})}{\Gamma(\frac12+\lambda+it)}. \end{aligned} \tag{6.42}$$

The endpoint $x = 0$ gives $\Re(C_1 + s) < 1/2$, while infinity, the ρ-sum, and the m-sum give respectively $\Re z_{\mathrm{d}} > 0$, $\Re z_{\mathrm{d}} > 1$, and $\Re\nu_{\mathrm{d}} > 1$. Hence

$$\max\{1 - \Re(\lambda + C_1), \Re(\lambda - B)\} < \Re s < \frac{1}{2} - \Re C_1.$$

The first lower bound is below the upper bound precisely when $\Re\lambda > 1/2$; the second is below it precisely when $\Re\lambda < 1/2 + \Re(B - C_1)$. This proves (6.32), including the necessity of its corridor.

The dual AFE piece makes the substitution $(B, C_1) \mapsto (-C_1, -B)$ and has the fixed exterior factor $X_{B,C_1}(t)$. Its positive-ρ integral is, explicitly,

$$I_{\mathrm{q},+}(\mathfrak{r}) := \int_0^\infty y^{-1/2-\lambda-it}(y+\mathfrak{r})^{-1/2+B+it-s}\,\mathrm{d}y$$

$$= \mathfrak{r}^{-z_{\mathrm{q}}} \frac{\Gamma(\frac{1}{2}-\lambda-it)\Gamma(z_{\mathrm{q}})}{\Gamma(\frac{1}{2}-B-it+s)}. \tag{6.43}$$

The $y = 0$ condition is $\Re\lambda < 1/2$; the infinity, ρ-, and m-conditions are respectively $\Re z_{\mathrm{q}} > 0$, $\Re z_{\mathrm{q}} > 1$, and $\Re\nu_{\mathrm{q}} > 1$. They give (6.33).

Finally, the dual, negative-ρ branch is

$$\begin{aligned} I_{\mathrm{q},-}(\mathfrak{r}) &:= \int_0^\infty (x+\mathfrak{r})^{-1/2-\lambda-it} x^{-1/2+B+it-s}\,\mathrm{d}x \\ &= \mathfrak{r}^{-z_{\mathrm{q}}} \frac{\Gamma(\frac{1}{2}+B+it-s)\Gamma(z_{\mathrm{q}})}{\Gamma(\frac{1}{2}+\lambda+it)}. \end{aligned} \tag{6.44}$$

Here $x = 0$ requires $\Re s < 1/2 + \Re B$; infinity and the two sums require $\Re z_{\mathrm{q}} > 1$ and $\Re\nu_{\mathrm{q}} > 1$. Therefore

$$\max\{1-\Re(\lambda-B), \Re(\lambda+C_1)\} < \Re s < \frac{1}{2} + \Re B.$$

The two comparisons with the upper endpoint give separately $\Re\lambda > 1/2$ and $\Re\lambda < 1/2 + \Re(B - C_1)$, proving (6.34). Since $\Re(B - C_1) > 0$ on the good patch, both negative intervals are nonempty.

We next justify the contour operation before evaluating any beta integral. Keep R, κ fixed and truncate the complete s- and u-lines at a common height V. At this finite-regulator stage the possible singularities of the unreduced source Mellin integrand are as follows.

(a) The pole $s = 0$ comes from $1/s$. Every source and destination s-line just used has positive real part, so it is not crossed.

(b) The poles $u = 0, -1, -2, \ldots$ come from $\Gamma(u)$. The line $\Re u = c_u > 0$ is not moved, so none is crossed.

(c) The numerator-gamma poles of g_{B,C_1} are

$$s = -\frac{1}{2} - B - it - 2j, \qquad s = -\frac{1}{2} - C_1 + it - 2j \quad (j \geq 0),$$

and those of $g_{-C_1,-B}$ are

$$s = -\frac{1}{2} + C_1 - it - 2j, \qquad s = -\frac{1}{2} + B + it - 2j \quad (j \geq 0).$$

For large T their real parts lie to the left of every positive line used above. Denominator gamma factors give zeros, not poles.

(d) The kernel $\mathcal{G}(s,u)$ is entire in (s,u); its displayed denominators depend only on the external shifts and stay bounded away from zero on the good patch. Because $m, |\rho|$ are finite at this stage, there is no arithmetic zeta function and hence no $z = 1$ or $\nu = 1$ pole.

Thus the finite rectangles deform without a residue. Let $\mathcal{I}^{R,\kappa}_{\epsilon,\sigma}(s,u,t)$ denote the finite m, ρ sum of the integrand in (6.39), with $w(t)$ and the differentials omitted. Cauchy's theorem gives the exact finite-V identity

$$\mathscr{S}^{R,\kappa,V}_{\epsilon,\sigma} - \mathscr{T}^{R,\kappa,V}_{\epsilon,\sigma} = \mathscr{H}^{R,\kappa,V}_{\epsilon,\sigma}, \tag{6.45}$$

where

$$\mathscr{H}_{\epsilon,\sigma}^{R,\kappa,V} = -\int_{\mathbb{R}} w(t)\frac{1}{(2\pi i)^2}\int_{[c_u]_V}\left\{\int_{\sigma_{\epsilon,\sigma}-iV}^{2-iV} + \int_{2+iV}^{\sigma_{\epsilon,\sigma}+iV}\right\}\mathcal{I}_{\epsilon,\sigma}^{R,\kappa}(s,u,t)\,\mathrm{d}s\,\mathrm{d}u\,\mathrm{d}t. \tag{6.46}$$

The two horizontal integrals carry the displayed orientations. On either horizontal s-edge, $s = \sigma + iV$,

$$|e^{s^2}| = e^{\sigma^2 - V^2},$$

while the gamma ratios and remaining factors have at most exponential times polynomial growth in V. Hence $\mathscr{H}_{\epsilon,\sigma}^{R,\kappa,V} \to 0$ for fixed R, κ, locally uniformly in A. No u-contour is moved in this proposition. After the complete s-lines have been restored, the vertical u-tails are controlled by Stirling:

$$|\Gamma(c_u + iV)| \ll (1+V)^{c_u - 1/2}e^{-\pi V/2}.$$

All other u-dependence is polynomial at fixed R, κ. Consequently, after $V \to \infty$ in (6.45), the two complete regulated vertical-contour integrals agree exactly. This is the first limit in (6.29); no variable regulator has yet been removed.

We now give the uniform dominators for the remaining two limits. Let

$$\mathscr{K}_A \Subset \{\Re A < \tfrac{1}{2} - c_u/2\}, \qquad \mathscr{K}_g \Subset \Omega_g, \tag{6.47}$$

and let $a^{\pm}$ be the supremum and infimum of $\Re A$. On $\mathscr{K}_g$ put

$$\begin{aligned} b_* &:= \inf \min\{\Re B, -\Re C_1\} > 0,\\ c^+ &:= \sup \max\{\Re C_1, -\Re B\},\\ c^- &:= \inf \min\{\Re C_1, -\Re B\}. \end{aligned}$$

These choices simultaneously cover the direct and dual substitutions. Put

$$\omega_A(l) = \begin{cases} l^{-1/2-a^+}, & 0 < l \le 1,\\ l^{-1/2-a^-}, & l > 1, \end{cases} \qquad \omega_C(n) = \begin{cases} n^{-1/2-c^+}, & 0 < n \le 1,\\ n^{-1/2-c^-}, & n > 1. \end{cases}$$

After the t-integral has been performed, choose P, Q, J once, depending only on the distances of (6.47) from their boundaries. The four source branches and any prescribed finite number of A-derivatives are dominated, up to a fixed logarithmic power and the finite outer h, k weights, by the single nonnegative function

$$\mathfrak{M}_{\mathrm{src}}(l,m,n) := T\,\omega_A(l)m^{-1/2-b_*}\omega_C(n)\left(1+\frac{l}{Y}\right)^{-P}\left(1+\frac{mn}{T}\right)^{-Q} \times \left\{\mathbf{1}_{\{l,n\ge 1/2\}} + \mathbf{1}_{\{\min(l,n)<1\}}\left(1+\Delta\left|\log\frac{hn}{klm}\right|\right)^{-J}\right\}. \tag{6.48}$$

Logarithms inserted by A-derivatives are absorbed by increasing P, Q, J. The mesh estimate (6.17), the product bound (6.18), and the endpoint estimates (6.20)–(6.25) show that the sum–integral of (6.48) is finite, uniformly for $(A, B, C_1) \in \mathscr{K}_A \times \mathscr{K}_g$. Since

$$\mathbf{1}_{\{R^{-1}\le l,n\le R\}}\mathbf{1}_{\{m,\rho\le R\}}e^{-\kappa(m+\rho+l+n)} \le 1,$$

this majorant is independent of R and κ. It gives dominated convergence at the source, first as $R \to \infty$ with $\kappa > 0$ fixed and then as $\kappa \to 0^+$, locally uniformly on the stated A-compact sets.

For a destination, restrict temporarily to a compact subset $\mathscr{K}_{\mathrm{Ton}}$ of its branchwise Tonelli open set, and choose the line $\Re s = \sigma_{\epsilon,\sigma}$ with a common margin $\delta > 0$ such that

$$\Re z_\epsilon \ge 1 + \delta, \qquad \Re \nu_\epsilon \ge 1 + \delta, \tag{6.49}$$

and the relevant zero-endpoint inequality in (6.31)–(6.34) also has margin δ. After scaling v by $\mathfrak{r} = d\rho$, define

$$b_{\epsilon,+}(v) = \sup_{\mathscr{K}_{\mathrm{Ton}}} v^{-1/2-\Re\lambda}(1+v)^{-1/2-\Re C_\epsilon - \Re s},$$
$$b_{\epsilon,-}(v) = \sup_{\mathscr{K}_{\mathrm{Ton}}} v^{-1/2-\Re C_\epsilon-\Re s}(1+v)^{-1/2-\Re\lambda}.$$

The endpoint margins give $\int_0^\infty b_{\epsilon,\sigma}(v)\,\mathrm{d}v < \infty$. Stirling and the Gaussian factor in $\mathcal{G}$ give, on the complete contours,

$$C_{\mathscr{K}_{\mathrm{Ton}}}(1+|\tau_u|+|\tau_s|)^N e^{-\pi|\tau_u|/3-\tau_s^2/2} m^{-1-\delta}\rho^{-1-\delta} b_{\epsilon,\sigma}(v), \tag{6.50}$$

for some fixed N. Indeed the gcd factor after scaling is $d^{1-\Re z_\epsilon} \le 1$. Thus (6.50) is integrable in $(\tau_u, \tau_s, m, \rho, v)$ and is independent of R, κ. It justifies the same ordered limits at every destination, uniformly on $\mathscr{K}_{\mathrm{Ton}}$. Notice that destination domination is asserted only inside a Tonelli chamber; outside it, holomorphy comes from the source and the identity theorem, not from an unproved absolute majorant.

The recovery order is now strict. First restore the complete contours ($V \to \infty$), then remove the geometric truncation ($R \to \infty$) while $\kappa > 0$, and then remove the Abel factor ($\kappa \to 0^+$). Only after the last step are the unregulated beta integrals (6.41)–(6.44) evaluated. Their powers $\mathfrak{r}^{-z_\epsilon} = d^{-z_\epsilon}\rho^{-z_\epsilon}$ then permit, by absolute convergence in the Tonelli chamber, first the ρ-sum $\sum_{\rho\ge1}\rho^{-z_\epsilon} = \zeta(z_\epsilon)$ and then the m- and gcd-sum giving $C_{h,k}$. The zeta functional equation and the finite Euler identity of Lemma 7.1 are applied only after these series have been recovered. Thus no Abel-weighted integral is identified with a beta quotient, and no Abel-weighted arithmetic sum is identified with an Euler product.

For completeness, singularities appearing only after the variable regulators have been removed must be recorded separately. The direct branches produce $\zeta(z_{\mathrm{d}})$ and a gcd–m factor with polar part $\zeta(\nu_{\mathrm{d}})$; the dual branches similarly produce $\zeta(z_{\mathrm{q}})$ and $\zeta(\nu_{\mathrm{q}})$.

(i) The direct pole $z_{\rm d} = 1$ is cancelled by the kernel zero $s = 1 - \lambda - C_1$, and the dual pole $z_{\rm q} = 1$ by the zero $s = 1 - \lambda + B$.

(ii) After applying the zeta functional equation, the direct hyperplanes $z_{\rm d} = 0$, $\nu_{\rm d} = 1$, and the corresponding dual hyperplanes $z_{\rm q} = 0$, $\nu_{\rm q} = 1$ remain. They are not discarded or crossed branchwise: these are the small arithmetic divisors cancelled only after the four branches and both diagonals are assembled.

(iii) Before the positive- and negative-ρ pieces are combined, the beta factors may have the following gamma poles, where $j \geq 0$:

$$\begin{aligned}
(\mathrm{d},+):&\quad z_{\rm d} = -j, \quad \lambda = \frac{1}{2} + j - it;\\
(\mathrm{d},-):&\quad z_{\rm d} = -j, \quad s = \frac{1}{2} - C_1 + it + j;\\
(\mathrm{q},+):&\quad z_{\rm q} = -j, \quad \lambda = \frac{1}{2} + j - it;\\
(\mathrm{q},-):&\quad z_{\rm q} = -j, \quad s = \frac{1}{2} + B + it + j.
\end{aligned}$$

They lie outside the four Tonelli domains. After combining the two ρ-signs, Lemma 7.2 cancels every apparent near-real archimedean pole. The remaining direct poles are

$$s = n - \frac{1}{2} - C_1 + it \ (n \geq 1), \qquad s = -\frac{1}{2} - B - it - 2j, \qquad s = -\frac{1}{2} - C_1 + it - 2j,$$

and the remaining dual poles are

$$s = n - \frac{1}{2} + B + it \ (n \geq 1), \qquad s = -\frac{1}{2} + C_1 - it - 2j, \qquad s = -\frac{1}{2} + B + it - 2j.$$

All have imaginary height comparable with T. If a later fixed-line deformation crosses one, e^{s^2} makes its residue smaller than every power of T. Reciprocal gamma factors again give zeros only.

The poles $s = 0$ and those of $\Gamma(u)$ have not moved during this bridge; they are treated in the subsequent fixed-u assembly and final u-move.

It remains to identify the selected continuation. The positive branches agree with their destinations on the nonempty open set

$$\mathscr{U}_+ = \{A : \Re A + c_u < 1/2\} \cap \{\Re A < 1/2 - c_u/2\},$$

whereas the negative branches agree on

$$\mathscr{U}_- = \left\{A : \frac{1}{2} < \Re A + c_u < \frac{1}{2} + \Re(B - C_1)\right\} \cap \{\Re A < 1/2 - c_u/2\}.$$

The latter is nonempty because $\Re(B - C_1) > 0$; for large T its width is $O(1/\log T) < c_u/2$, so it lies in the connected strip (6.16). Apply the one-variable identity theorem separately to each complete, already integrated branch as a function of A. This extends an exact equality to the target A; it does not propagate the endpoint-deletion $O(T^{-N})$ error and does not treat λ and s as independent continuation variables. $\square$

7. Beta integrals, Euler factors, and gamma identities

We now evaluate the endpoint-deleted zero mode at the Tonelli destinations constructed in Section 6. Put

$$\lambda = \alpha + u, \qquad z = \lambda + \gamma + s, \qquad q_1 = 1 - z, \qquad v_1 = 1 - \lambda + \beta + s. \tag{7.1}$$

Every expression in this section is the fully unregulated destination limit of Proposition 6.4: the complete s, u contours have already been restored, R has gone to infinity, and the Abel parameter κ has already gone to zero. We first evaluate the ordinary beta integral, then recover the absolutely convergent ρ-series, and only then recover the m- and gcd-series. Thus none of the beta or Euler identities below is applied to an Abel-weighted expression.

For $r > 0$, ordinary beta integration gives

$$\begin{aligned}\int_0^\infty y^{-1/2-\lambda-it}(y+r)^{-1/2-\gamma+it-s}\,\mathrm{d}y \\ = r^{-z}\frac{\Gamma(\frac{1}{2}-\lambda-it)\Gamma(z)}{\Gamma(\frac{1}{2}+\gamma-it+s)}.\end{aligned} \tag{7.2}$$

A genuine absolute-convergence domain, including the subsequent ρ- and m-sums, is

$$\Re\lambda < \frac{1}{2}, \qquad \Re z > 1, \qquad \Re v_1 > 1. \tag{7.3}$$

For $r = -R < 0$, substitute $y = R + x$ to obtain

$$\begin{aligned}\int_0^\infty (x+R)^{-1/2-\lambda-it}x^{-1/2-\gamma+it-s}\,\mathrm{d}x \\ = R^{-z}\frac{\Gamma(\frac{1}{2}-\gamma+it-s)\Gamma(z)}{\Gamma(\frac{1}{2}+\lambda+it)}.\end{aligned} \tag{7.4}$$

Here a genuine domain is

$$\Re(\gamma+s) < \frac{1}{2}, \qquad \Re z > 1, \qquad \Re v_1 > 1. \tag{7.5}$$

The two signs need not have a common elementary chamber near the intended shifts; Section 6 proves the two identities separately and then selects their common meromorphic continuation by regulated exact equality.

The sum over ρ supplies $\zeta(z)$. We use

$$\zeta(z) = \chi(z)\zeta(1-z), \qquad \chi(z) = \pi^{z-1/2}\frac{\Gamma((1-z)/2)}{\Gamma(z/2)}. \tag{7.6}$$

The remaining gcd sum is

$$C_{h,k}(s,u) = \sum_{m\ge 1}\frac{(h,km)^{q_1}}{m^{v_1}}. \tag{7.7}$$

Lemma 7.1 (Finite Euler factors)**.** *In the common meromorphic continuation,*

$$h^{-1/2+\gamma+s}k^{-1/2+\lambda}\zeta(1-z)C_{h,k}(s,u) = Z_{-\gamma-s,\,\beta+s,\,-\lambda}(h,k). \tag{7.8}$$

Proof. Write $g = (h,k)$, $h = gh_0$, $k = gk_0$. Since $(h, km) = g(h_0, m)$,

$$C_{h,k} = g^{q_1}\zeta(v_1)\prod_{p^\nu \| h_0}(1-p^{-v_1})\sum_{j\geq 0} p^{q_1\min(j,\nu)-v_1 j}. \tag{7.9}$$

For $X = p^{-q_1}$ and $Y_0 = p^{-v_1}$, splitting the sum at $j = \nu$ gives

$$(1-Y_0)\sum_{j\geq 0} X^{-\min(j,\nu)}Y_0^j = X^{-\nu}(1-X)(1-Y_0)\sum_{i+j\geq \nu} X^i Y_0^j. \tag{7.10}$$

This is exactly the local factor in (2.5), after the exterior powers in (7.8) are restored. ☐

Combine the two signs in the archimedean factor

$$\Phi_{\lambda;\beta,\gamma}(s,t) = \chi(z)\Gamma(z)g_{\beta,\gamma}(s,t)\left\{\frac{\Gamma(\frac{1}{2}-\lambda-it)}{\Gamma(\frac{1}{2}+\gamma-it+s)} + \frac{\Gamma(\frac{1}{2}-\gamma+it-s)}{\Gamma(\frac{1}{2}+\lambda+it)}\right\}. \tag{7.11}$$

There is no additional π^{-s} in (7.11), because it is already part of $g_{\beta,\gamma}$.

Lemma 7.2 (Archimedean pole collapse)**.** *Let $a_0 = \frac{1}{2} - \lambda - it$. Then*

$$\chi(z)\Gamma(z)\left\{\frac{\Gamma(a_0)}{\Gamma(a_0+z)} + \frac{\Gamma(1-a_0-z)}{\Gamma(1-a_0)}\right\} = (2\pi)^z\frac{\Gamma(a_0)}{\Gamma(a_0+z)}\frac{\sin\pi(a_0+z/2)}{\sin\pi(a_0+z)}. \tag{7.12}$$

For $t \asymp T$ and $|\Im\lambda| + |\Im s| \leq (\log T)^2$, the left side has no pole near the real s-axis. Its possible poles occur at

$$s = n - \frac{1}{2} - \gamma + it \quad (n \geq 1), \tag{7.13}$$

and the numerator gamma factors of g add only

$$s = -\frac{1}{2} - \beta - it - 2j, \qquad s = -\frac{1}{2} - \gamma + it - 2j \quad (j \geq 0). \tag{7.14}$$

All are at imaginary height comparable with T.

Proof. Duplication and reflection give

$$\chi(z)\Gamma(z) = \frac{(2\pi)^z}{2\cos(\pi z/2)}$$

and

$$\frac{\Gamma(1-a_0-z)}{\Gamma(1-a_0)} = \frac{\Gamma(a_0)}{\Gamma(a_0+z)}\frac{\sin\pi a_0}{\sin\pi(a_0+z)}.$$

Now use

$$\sin\pi(a_0+z) + \sin\pi a_0 = 2\sin\pi(a_0+z/2)\cos(\pi z/2).$$

This proves (7.12), including cancellation of the apparent odd-integer poles of $1/\cos(\pi z/2)$. At nonpositive integral values of $a_0 + z$, the reciprocal gamma zero cancels the sine denominator. The remaining poles are (7.13) and (7.14). If such a pole is crossed between fixed vertical lines, the factor e^{s^2} makes its residue $O_D(T^{-D})$. $\square$

Lemma 7.3 (Archimedean reflection)**.** *As meromorphic identities,*

$$\Phi_{\lambda;\beta,\gamma}(0,t) = X_{\lambda,\gamma}(t), \tag{7.15}$$

and

$$X_{\beta,\gamma}(t)\Phi_{\lambda;-\gamma,-\beta}(s,t) = \Phi_{\lambda;\beta,\gamma}(-s,t). \tag{7.16}$$

Proof. Substitute (2.6), (7.6), and (7.11). Pair the full gamma factors by

$$\Gamma(z)\Gamma(1-z) = \frac{\pi}{\sin \pi z}$$

and the half-arguments by

$$\Gamma(z)\Gamma(z+\tfrac{1}{2}) = 2^{1-2z}\sqrt{\pi}\,\Gamma(2z).$$

For (7.15), the two terms in braces combine by reflection and the remaining half-gamma factors form $X_{\lambda,\gamma}$. For (7.16), first combine the β,γ factors with $X_{\beta,\gamma}$ and then apply duplication. The identities hold on a pole-free open set and hence everywhere by meromorphic continuation. $\square$

The endpoint-deleted direct and dual zero modes are therefore

$$\begin{aligned}\widetilde{\mathcal{O}}^{+}_{0,Y}(h,k) = {}& \int_{\mathbb{R}} w(t)\frac{1}{2\pi i}\int_{(c_u)} \Gamma(u)Y^u \frac{1}{2\pi i}\int_{(c_s)} \frac{\mathcal{G}(s,u)}{s}\Phi_{\lambda;\beta,\gamma}(s,t) \\ &\times Z_{-\gamma-s,\beta+s,-\lambda}(h,k)\,\mathrm{d}s\,\mathrm{d}u\,\mathrm{d}t, \end{aligned}\tag{7.17}$$

$$\begin{aligned}\widetilde{\mathcal{O}}^{-}_{0,Y}(h,k) = {}& \int_{\mathbb{R}} w(t)\frac{1}{2\pi i}\int_{(c_u)} \Gamma(u)Y^u \frac{1}{2\pi i}\int_{(c_s)} \frac{\mathcal{G}(s,u)}{s}X_{\beta,\gamma}(t)\Phi_{\lambda;-\gamma,-\beta}(s,t) \\ &\times Z_{\beta-s,-\gamma+s,-\lambda}(h,k)\,\mathrm{d}s\,\mathrm{d}u\,\mathrm{d}t. \end{aligned}\tag{7.18}$$

The line $c_s > 0$ is first chosen in the branchwise absolute-convergence regions; elsewhere these formulas denote the continuation selected by the full-kernel bridge of Section 6.

8. Fixed-u residue assembly and quantitative continuation

For $\lambda = \alpha + u$, define the complete three-permutation expression

$$\begin{aligned}\mathcal{P}_\lambda(h,k;t) = {}& Z_{\lambda,\beta,\gamma}(h,k) + X_{\lambda,\gamma}(t)Z_{-\gamma,\beta,-\lambda}(h,k) \\ &+ X_{\beta,\gamma}(t)Z_{\lambda,-\gamma,-\beta}(h,k). \end{aligned}\tag{8.1}$$

The individual terms are meromorphic. They will always be combined before a collision divisor is crossed.

The direct and dual diagonal terms for fixed h,k are

$$\begin{aligned}\mathcal{D}^{+}(h,k) = {}& \int_{\mathbb{R}} w(t)\frac{1}{2\pi i}\int_{(c_u)} \Gamma(u)Y^u \\ &\times \frac{1}{2\pi i}\int_{(c_s)} \frac{\mathcal{G}(s,u)}{s} g_{\beta,\gamma}(s,t)Z_{\lambda,\beta+s,\gamma+s}(h,k)\,\mathrm{d}s\,\mathrm{d}u\,\mathrm{d}t, \end{aligned}\tag{8.2}$$

$$\mathcal{D}^-(h,k) = \int_{\mathbb{R}} w(t) \frac{1}{2\pi i} \int_{(c_u)} \Gamma(u) Y^u X_{\beta,\gamma}(t) \times \frac{1}{2\pi i} \int_{(c_s)} \frac{\mathcal{G}(s,u)}{s} g_{-\gamma,-\beta}(s,t) Z_{\lambda,-\gamma+s,-\beta+s}(h,k) \,\mathrm{d}s\,\mathrm{d}u\,\mathrm{d}t. \tag{8.3}$$

8.1. **The three residue cancellations.** Reflect the dual diagonal and dual zero mode by $s \mapsto -s$. Before any term is separated, the small arithmetic poles at fixed u are

term	pole locations before reflection where indicated
$\mathcal{D}^+$	$s = -\lambda - \gamma$, $s = -(\beta+\gamma)/2$
$\mathcal{D}^-$	$s = \beta - \lambda$, $s = (\beta+\gamma)/2$
zero mode	$s = -\lambda - \gamma$, $s = \lambda - \beta$

The exact identities required at these poles are

$$X_{\beta,\gamma}(t) g_{-\gamma,-\beta}(-s,t) = g_{\beta,\gamma}(s,t) X_{\beta+s,\gamma+s}(t), \tag{8.4}$$

$$\Phi_{\lambda;\beta,\gamma}(-\lambda-\gamma,t) = g_{\beta,\gamma}(-\lambda-\gamma,t), \tag{8.5}$$

$$\Phi_{\lambda;\beta,\gamma}(\lambda-\beta,t) = g_{\beta,\gamma}(\lambda-\beta,t) X_{\lambda,\gamma+\lambda-\beta}(t). \tag{8.6}$$

They follow from (2.6), (7.11), reflection and duplication.

At $s = -\lambda-\gamma$, the direct diagonal and zero-mode arithmetic factors both equal

$$Z_{\lambda,\beta-\lambda-\gamma,-\lambda}(h,k), \tag{8.7}$$

while their polar zeta arguments have derivatives $+1$ and -1. Equation (8.5) therefore cancels the residues. After reflection, the second pair at $s = \lambda - \beta$ has common factor

$$Z_{-\gamma-\lambda+\beta,\lambda,-\lambda}(h,k), \tag{8.8}$$

and (8.6) gives opposite residues. Finally, at $s_0 = -(\beta+\gamma)/2$, the two diagonal factors specialize to

$$Z_{\lambda,(\beta-\gamma)/2,(\gamma-\beta)/2}(h,k). \tag{8.9}$$

Their polar arguments have derivatives $+2$ and -2, and $X_{(\beta-\gamma)/2,(\gamma-\beta)/2}(t) = 1$. Thus the third pair also cancels. The equalities in (8.7)–(8.9) are equalities of the complete Euler products, not just their global zeta factors.

Choose

$$\sigma_s = -\frac{1}{2} + \varepsilon_s, \qquad 0 < \varepsilon_s < \min\left(\frac{1}{4}, \frac{1}{2} - c_u\right), \tag{8.10}$$

and let C_* bound the scaled real parts of the shifts. For large T, take

$$c_u + \frac{C_*}{\log T} < c_s < \min\left(\frac{1}{2} - \varepsilon_s, 1 - c_u - \frac{C_*}{\log T}\right). \tag{8.11}$$

In the central strip $-c_s < \Re s < c_s$, the complete reflected integrand has only

$$\begin{array}{c|c} x = s + \lambda + \gamma = 0 & s = -\lambda - \gamma, \\ y = s - \lambda + \beta = 0 & s = \lambda - \beta, \\ z = 2s + \beta + \gamma = x + y = 0 & s = -(\beta + \gamma)/2, \\ s = 0 & s = 0. \end{array} \tag{8.12}$$

The $z = 1$ locations lie outside by (8.11) and are zeros of $\mathcal{G}$ in any wider deformation. By Lemma 7.2, all other archimedean poles have imaginary part comparable with t and are Gaussian-small.

Put

$$F_0(s) = \mathcal{G}(s,u)\Phi_{\lambda;\beta,\gamma}(s,t)Z_{-\gamma-s,\beta+s,-\lambda}(h,k). \tag{8.13}$$

Using (7.16), evenness of $\mathcal{G}$, and symmetry of Z in its first two shifts, the dual zero mode becomes exactly

$$-\frac{1}{2\pi i}\int_{(-c_s)} \frac{F_0(s)}{s}\,\mathrm{d}s. \tag{8.14}$$

The sign comes from $1/(-s)$; reversal of the parametrized contour introduces no second sign. Thus the two zero modes form a contour difference, evaluated by the $x = 0$, $y = 0$, and $s = 0$ residues, with no zero-mode boundary line.

Move the direct diagonal from (c_s) to (σ_s) and the reflected dual diagonal from $(-c_s)$ to $(-\sigma_s)$. The three nonzero residue pairs cancel as above. The $s = 0$ residues of the two diagonals and the zero mode are precisely the three terms in $Q(u)\mathcal{P}_\lambda$. Undoing the reflection on the final dual line leaves the full diagonal boundary

$$\begin{aligned} \mathcal{R}_{\mathrm{diag},Y}(h,k) = \int_{\mathbb{R}} w(t)\frac{1}{2\pi i}\int_{(c_u)} &\Gamma(u)Y^u \\ \times \Bigg\{ &\frac{1}{2\pi i}\int_{(\sigma_s)} \frac{\mathcal{G}(s,u)}{s} g_{\beta,\gamma}(s,t) \\ &\qquad \times Z_{\alpha+u,\beta+s,\gamma+s}(h,k)\,\mathrm{d}s \\ &+ X_{\beta,\gamma}(t)\frac{1}{2\pi i}\int_{(\sigma_s)} \frac{\mathcal{G}(s,u)}{s} g_{-\gamma,-\beta}(s,t) \\ &\qquad \times Z_{\alpha+u,-\gamma+s,-\beta+s}(h,k)\,\mathrm{d}s\Bigg\}\,\mathrm{d}u\,\mathrm{d}t. \end{aligned} \tag{8.15}$$

The two terms in (8.15) are separated only after the joint cancellation at the three small divisors.

Combining Proposition 6.2 with this fixed-u residue calculation yields, on the good patch and after the outer coefficient sum,

$$\begin{aligned} &\mathcal{D}^+ + \mathcal{D}^- + \mathcal{O}^+_{0,Y} + \mathcal{O}^-_{0,Y} \\ &\qquad = \int_{\mathbb{R}} w(t)\frac{1}{2\pi i}\int_{(c_u)} \Gamma(u)Y^u Q(u)\mathcal{P}_{\alpha+u}(h,k;t)\,\mathrm{d}u\,\mathrm{d}t \end{aligned}$$

$$+\,\mathcal{R}_{\mathrm{diag},Y}(h,k)+O_N(T^{-N}) \quad (8.16)$$

for every $N>0$. As usual in an h,k identity, the error in (8.16) is understood after summation with $a_h\overline{b_k}/\sqrt{hk}$.

8.2. **Quantitative continuation of the assembled error.** For external shifts A,B,C_1, let $\mathscr{A}_T(A,B,C_1)$ denote the complete source: both diagonals and both exact zero modes, with their endpoint cutoffs, common positive u-line, and outer coefficient sum. Let

$$\begin{aligned}\mathscr{R}_T(A,B,C_1) = \sum_{h\le H}\sum_{k\le K}\frac{a_h\overline{b_k}}{\sqrt{hk}} \\ \times\int_{\mathbb{R}} w(t)\frac{1}{2\pi i}\int_{(c_u)}\Gamma(u)Y^uQ(u)\mathcal{P}_{A+u}(h,k;t)\,\mathrm{d}u\,\mathrm{d}t,\end{aligned} \quad (8.17)$$

$$\mathscr{D}_T(A,B,C_1) = \sum_{h\le H}\sum_{k\le K}\frac{a_h\overline{b_k}}{\sqrt{hk}}\mathcal{R}_{\mathrm{diag},Y}(h,k). \quad (8.18)$$

The shifts in every displayed kernel are changed from (α,β,γ) to (A,B,C_1).

Lemma 8.1 (Rapid-decay propagation)**.** *Let $\Omega\subset\mathbb{C}^n$ be a fixed connected domain, let $U\Subset\Omega$ be nonempty and open, and let $K_0\Subset\Omega$. If f_T is holomorphic on Ω and*

$$\sup_{\Omega}|f_T|\le T^{C_2},\qquad \sup_{U}|f_T|\ll_N T^{-N}\quad \textit{for every } N, \quad (8.19)$$

then, for every $D>0$,

$$\sup_{K_0}|f_T|\ll_{D,K_0} T^{-D}. \quad (8.20)$$

Proof. Join a ball compactly contained in U to K_0 by finitely many overlapping balls in Ω. Repeated use of the three-ball, or two-constants, inequality gives

$$|f_T|\le (T^{-N})^{\omega}(T^{C_2})^{1-\omega}$$

on K_0, where $\omega>0$ depends only on the fixed geometry. Take $N>(D+(1-\omega)C_2)/\omega$. □

Dependency note (logical order). The collision-removability result Proposition 9.3, although placed with the collision discussion in Section 9, is an algebraic input to the next proposition, not a consequence of it. More precisely, its proof uses only the explicit three-term definition (8.1), the Euler product and archimedean identities (2.5) and (2.6), and the local division statement Lemma 9.2. It uses neither Lemma 8.1 nor Proposition 8.2, and it uses no polynomial majorant or contour move proved below. Thus the logical dependency chain is

$$\begin{gathered}(8.1),\ (2.5),\ (2.6)\\ \Downarrow\\ \textit{Lemma } 9.2\\ \Downarrow\\ \textit{Proposition } 9.3\\ \Downarrow\\ \textit{Proposition } 8.2.\end{gathered}$$

the later placement of the middle two statements is expository only.

Proposition 8.2 (Full-kernel assembly continuation)**.** *Under* (2.8)*, for every fixed* $C_0, D > 0$*,*

$$\mathscr{A}_T(A, B, C_1) - \mathscr{R}_T(A, B, C_1) - \mathscr{D}_T(A, B, C_1) = O_D(T^{-D}) \tag{8.21}$$

uniformly for

$$|A| + |B| + |C_1| \le C_0/\log T, \tag{8.22}$$

including every shift collision.

Proof. Put $L = \log T$ and use the scaled variables

$$(\mathsf{a}, \mathsf{b}, \mathsf{c}) = L(A, B, C_1). \tag{8.23}$$

We first fix all geometry. The admissible contour choices

$$c_u = \frac{1}{8}, \qquad \varepsilon_s = \frac{1}{8}, \qquad \sigma_s = -\frac{3}{8}, \qquad c_s = \frac{1}{4} \tag{8.24}$$

satisfy (3.10), (8.10), and, for sufficiently large T, (8.11). For the prescribed C_0, set

$$\begin{aligned} R_* &= C_0 + 6,\\ \Omega &= \{(\mathsf{a}, \mathsf{b}, \mathsf{c}) : |\mathsf{a}|, |\mathsf{b}|, |\mathsf{c}| < R_*\},\\ \Omega^\sharp &= \{(\mathsf{a}, \mathsf{b}, \mathsf{c}) : |\mathsf{a}|, |\mathsf{b}|, |\mathsf{c}| < R_* + 1\}. \end{aligned} \tag{8.25}$$

The target compact set

$$K_0 = \{(\mathsf{a}, \mathsf{b}, \mathsf{c}) : |\mathsf{a}| + |\mathsf{b}| + |\mathsf{c}| \le C_0\} \tag{8.26}$$

has a fixed collar in Ω. We use the fixed open good-patch subset

$$U = \left\{ |\mathsf{a}| < \frac{1}{8}, |\mathsf{b} - 2| < \frac{1}{8}, |\mathsf{c} + 2| < \frac{1}{8} \right\}. \tag{8.27}$$

Its closure lies in Ω, and U lies in (6.15) with $\kappa_0 = 1$ and $c_1 = 5$, since on U one has $\Re\mathsf{b} > 15/8$, $\Re\mathsf{c} < -15/8$, and $|\mathsf{a}| + |\mathsf{b}| + |\mathsf{c}| < 35/8$.

Take T large enough that

$$L > 64(R_* + 1). \tag{8.28}$$

On $\Omega^\sharp$ the two linear denominator factors in (3.2) then have modulus at least $7/8$, and hence their squares have modulus at least $49/64$. With $C_* =$

$2(R_*+1)$, (8.24) satisfies (8.11). These are fixed contours throughout $\Omega^\sharp$; the unused kernel divisors stay a fixed positive distance away.

We now prove a polynomial majorant separately for each assembled object. In the convention of (2.2)– (2.3), all constants below may depend on R_*, B_0, M, ε, on the coefficient-bound constants, and on the indicated finite seminorms, but not on H, K, T or the point of the scaled shift domain. No new weight-function dependence is introduced in this section. Choose the auxiliary divisor-bound exponent sufficiently small. Once M is fixed, all factors $L^{C_M}T^{O(\varepsilon)}$ are absorbed into one additional power of T. This last coarse absorption is used only for the polynomial majorant needed by quantitative continuation, not for the sharp T^ε in (2.11).

For the exact-cutoff zero-mode part of $\mathscr{A}_T$, the mesh bound (6.17) and the full-kernel product bound give (6.18). Uniformly on $\Omega^\sharp$,

$$\int_{1/2}^{\infty} l^{-1/2-\Re A}(1+l/Y)^{-P}\,\mathrm{d}l \ll_{R_*,B_0} Y^{1/2+(R_*+1)/L} \ll T^{B_0/2}, \tag{8.29}$$

$$\sum_{m\geq 1} m^{-1/2-\Re B}\Big\{ \int_{1/2}^{\infty} n^{-1/2-\Re C_1}(1+mn/T)^{-Q}\,\mathrm{d}n + (1+m/T)^{-Q}\Big\} \ll_{R_*} T^{1/2}L. \tag{8.30}$$

Here P, Q are fixed sufficiently large product-decay orders. To see the second estimate, split at $m = T$. For $m \leq T$ the integral term is at most

$$T^{1/2-\Re C_1}\sum_{m\leq T} m^{-1-\Re B+\Re C_1} \ll_{R_*} T^{1/2}L,$$

because $\sum_{m\leq T} m^{-1+2(R_*+1)/L} \ll e^{2(R_*+1)}L$; increasing P, Q gives arbitrary convergence for $m > T$. This is an exact-cutoff source bound, not a claim of endpoint-deleted normal convergence in the opposite half-space.

The outside coefficients cost at most

$$\sum_{h\leq H}\sum_{k\leq K}\frac{|a_h b_k|}{\sqrt{hk}} \ll_\varepsilon T^\varepsilon (HK)^{1/2} \ll T^{1/2+\varepsilon}, \tag{8.31}$$

where $HK \leq T^{1-\eta}$ follows from (2.8) and $H \geq 1$. Together with the $O(T)$ length of the t-integral, (8.29)–(8.31) give $T^{B_0/2+2+o(1)}$ for each exact zero mode. On the positive line $\Re s = c_s = 1/4$, the two source diagonals are absolutely convergent and the Euler formula gives the deliberately cruder estimate

$$L^{C_M}T^{1+c_s+B_0c_u}H^{1+c_s+\varepsilon}K^{1/2+\varepsilon} \ll T^{B_0/8+4}. \tag{8.32}$$

The exact factors $X_{B,C_1}(t)$ are $O_{R_*}(1)$ on the scaled domain. The locally uniform exact-cutoff majorants prove holomorphy and

$$\sup_{\Omega^\sharp}|\mathscr{A}_T| \ll T^{B_0/2+4}. \tag{8.33}$$

For $\mathscr{R}_T$, retain the complete sum $\mathcal{P}_{A+u}$ before taking absolute values. On $\Re u = c_u = 1/8$, the collision coordinates

$$x = A + u + C_1, \qquad y = -B - C_1, \qquad z = A + u - B = x + y$$

satisfy $|x|, |z| \geq 3/32$ on $\Omega^\sharp$ by (8.28). Thus only $y = 0$ can meet this integration line. Away from a fixed scaled tube around $y = 0$, the Euler formula (2.5), Stirling, and vertical-strip zeta bounds give

$$\mathcal{P}_{A+u}(h, k; t) \ll_{R_*, M, \varepsilon} L^{C_M}(hk)^\varepsilon h^{1/2+\varepsilon}(1 + |\Im u|)^{C_M}. \tag{8.34}$$

In the scaled tube $|Ly| < 1/2$ we first use the logically prior algebraic input Proposition 9.3, with the acyclic dependency recorded above. More explicitly, fix A, C_1, u, t and put

$$f(w) = \mathcal{P}_{A+u}(h, k; t)|_{B=-C_1-w/L}\,.$$

The complete function f is holomorphic for $|w| < 1$ and the circle $|w| = 1/2$ lies inside the fixed collar of $\Omega^\sharp$. The maximum principle bounds every $f(w)$ with $|w| \leq 1/2$ by the maximum on this circle. There $|B + C_1| = 1/(2L)$, so the individual Euler estimates cost only a fixed power of L, while x and z remain bounded away from zero. A finite cover of $\overline{\Omega}$ proves (8.34) throughout the tube, including $y = 0$, without separately estimating a meromorphic term at the collision. The u, t, h, k integrations now give

$$\sup_{\Omega} |\mathscr{R}_T| \ll L^{C_M} T^{1+B_0/8} H^{1+\varepsilon} K^{1/2+\varepsilon} \ll T^{B_0/8+4}. \tag{8.35}$$

Finally consider $\mathscr{D}_T$. This object is defined only after the direct diagonal, reflected dual diagonal, and zero mode have been assembled and the three small residues have cancelled. On $\Re s = \sigma_s = -3/8$, the two pairs of Euler arguments have real parts

$$\begin{aligned}
\Re(1 + A + u + C_1 + s) &= 3/4 + O(R_*/L), \\
\Re(1 + B + C_1 + 2s) &= 1/4 + O(R_*/L), \\
\Re(1 + A + u - B + s) &= 3/4 + O(R_*/L), \\
\Re(1 - B - C_1 + 2s) &= 1/4 + O(R_*/L).
\end{aligned} \tag{8.36}$$

They stay a fixed distance from both zero and one. The prime-power formula (9.4) and vertical-strip zeta bounds therefore give, for both Euler products on this line,

$$Z \ll_{R_*, M, \varepsilon} L^{C_M}(hk)^{1/2+\varepsilon}(1 + |\Im u| + |\Im s|)^{C_M}. \tag{8.37}$$

Stirling gives $g(s, t) \ll T^{-3/8}(1 + |\Im s|)^{C_M}$; the factor e^{s^2} is Gaussian and $\Gamma(u)$ has exponential decay. Completing both vertical integrals and the h, k sums yields

$$\sup_{\Omega} |\mathscr{D}_T| \ll L^{C_M} T^{1-3/8+B_0/8} HK\, T^\varepsilon \ll T^{B_0/8+3}. \tag{8.38}$$

This estimate is for the assembled boundary object; no individual meromorphic beta branch has been continued through a collision.

It follows from (8.33), (8.35), and (8.38) that

$$F_T(\mathsf{a},\mathsf{b},\mathsf{c}) := \mathscr{E}_{\mathrm{end},T}(\mathsf{a}/L,\mathsf{b}/L,\mathsf{c}/L), \qquad \sup_{\Omega}|F_T| \ll T^{C_2}, \quad C_2 = \frac{B_0}{2}+5. \tag{8.39}$$

Thus the enlarged scaled domain, its collar, and the polynomial exponent are all fixed independently of T. On U, the good-patch assembly (8.16) gives $\sup_U |F_T| \ll_N T^{-N}$ for every fixed N.

The parameter order is acyclic. First fix $B_0 > 2$, C_0, and the desired final saving D; this fixes $K_0 \Subset \Omega$, $U \Subset \Omega$, and the propagation exponent $\omega = \omega(C_0) > 0$. Next choose one integer M such that, for $\sigma = M + 1/2$,

$$(B_0 - 1)\sigma > D + B_0 + 20. \tag{8.40}$$

This coarse condition also dominates the later permutation and diagonal-boundary left-line requirements. With this M fixed, $C_2 = B_0/2+5$ is fixed; only the lower threshold for T may depend on M, C_0. Now choose

$$N = \left\lceil \frac{D + (1-\omega)C_2}{\omega} \right\rceil + 2 \tag{8.41}$$

and afterward the endpoint integration-by-parts and product-tail orders needed for the good-patch $O(T^{-N})$. In particular M is independent of N. Applying Lemma 8.1 to F_T on the fixed geometry $U \Subset \Omega$, with target K_0, proves (8.21). □

Only the difference in (8.21) has been propagated. In particular, no power-sized estimate for an individual diagonal boundary term has been continued by unique continuation.

9. Final contour moves, collisions, and completion of the proof

The two u-moves in this section are performed only after Proposition 8.2. They are distinct: first we move the full diagonal boundary, and then the complete three-permutation expression.

9.1. **The diagonal-boundary u-move.**

The ordinary $u = 0$ residue of (8.15) is

$$\begin{aligned} \mathcal{R}_{\mathrm{diag},0}(h,k) = \int_{\mathbb{R}} w(t)\Bigg\{ &\frac{1}{2\pi i}\int_{(\sigma_s)} \frac{G(s)}{s} g_{\beta,\gamma}(s,t) \\ &\qquad\qquad \times Z_{\alpha,\beta+s,\gamma+s}(h,k)\,\mathrm{d}s \\ &+ X_{\beta,\gamma}(t)\frac{1}{2\pi i}\int_{(\sigma_s)} \frac{G(s)}{s} g_{-\gamma,-\beta}(s,t) \\ &\qquad\qquad \times Z_{\alpha,-\gamma+s,-\beta+s}(h,k)\,\mathrm{d}s\Bigg\}\,\mathrm{d}t. \end{aligned} \tag{9.1}$$

Move each u-line in (8.15) to

$$\Re u = -\sigma_u, \qquad \sigma_u = M + \frac{1}{2}. \tag{9.2}$$

On $\Re s = \sigma_s$, the only u-dependent arithmetic poles are

$$u = -\alpha - \gamma - s, \qquad u = \beta - \alpha - s. \tag{9.3}$$

Both have real part $1/2 - \varepsilon_s + O((\log T)^{-1}) > c_u$, so they lie to the right of the original line. The pole at $u = 0$ gives (9.1); the potential poles $u = -1, \ldots, -M$ vanish by (3.11).

For the new-line estimate, the finite Euler factor in (2.5) may be written

$$B_\nu(q, v) = X^\nu + (1 - X) \sum_{j=0}^{\nu-1} X^j Y_0^{\nu-j}, \qquad X = p^{-q}, \quad Y_0 = p^{-v}. \tag{9.4}$$

This follows by summing the geometric tail in

$$(1 - p^{-q})(1 - p^{-v}) \sum_{i+j\geq\nu} p^{-qi-vj}.$$

On (9.2), (9.4) and vertical-strip zeta bounds give

$$Z_{\alpha+u,\beta+s,\gamma+s}(h,k), \quad Z_{\alpha+u,-\gamma+s,-\beta+s}(h,k) \ll_M (\log T)^{C_M} (hk)^\varepsilon h^{\sigma_u+O(\varepsilon_s)+\varepsilon} \times (1 + |u| + |s|)^{C_M}. \tag{9.5}$$

Together with $g(s,t) \ll T^{\sigma_s}(1 + |s|)^{C_M}$, the gamma decay in u, and the Gaussian decay in s, the new lines after the outer coefficient sum are

$$\ll T^{1+\sigma_s+\varepsilon} Y^{-\sigma_u} H^{\sigma_u+O(\varepsilon_s)+\varepsilon} (HK)^{1/2}. \tag{9.6}$$

Since $H \leq T^{2(1-\eta)/3}$ and $Y = T^{B_0}$ with $B_0 > 2$, M can be chosen so that (9.6) is $O_D(T^{-D})$. At $u = 0$ the Euler arguments on $\Re s = \sigma_s$ remain a fixed distance from their poles, and hence

$$\sum_{h\leq H} \sum_{k\leq K} \frac{|a_h b_k|}{\sqrt{hk}} |\mathcal{R}_{\mathrm{diag},0}(h,k)| \ll_\varepsilon T^{1/2+\varepsilon} (HK)^{1/2}. \tag{9.7}$$

Consequently

$$\mathscr{D}_T = \sum_{h\leq H} \sum_{k\leq K} \frac{a_h \overline{b_k}}{\sqrt{hk}} \mathcal{R}_{\mathrm{diag},0}(h,k) + O_D(T^{-D}). \tag{9.8}$$

9.2. **The permutation u-move.**

Lemma 9.1 (Left-line bound). *For fixed $\sigma > 0$ and $\Re\lambda = -\sigma + O((\log T)^{-1})$,*

$$Q(u)\mathcal{P}_\lambda(h,k;t) \ll_{\sigma,M} (\log T)^{C_M} (hk)^\varepsilon (T^\sigma + h^\sigma)(1 + |\Im u|)^{C_M}. \tag{9.9}$$

After multiplication by $\Gamma(u)$, this is integrable on the complete vertical line.

Proof. Retain the complete sum $\mathcal{P}_\lambda$ before estimating it. In the collision coordinates

$$x = \lambda + \gamma, \qquad y = -\beta - \gamma, \qquad z = \lambda - \beta = x + y,$$

the line $\Re\lambda = -\sigma + O((\log T)^{-1})$ keeps x and z a fixed positive distance from zero. Thus only $y = 0$ can meet this line.

Away from $|y| < 1/(2\log T)$, use (9.4) term by term. If the first shift of Z is λ and its third shift is γ or $-\beta$, the local product is at worst $h_0^{\sigma+\varepsilon}$. If the third shift is $-\lambda$, both Euler arguments have real part $1 + \sigma + O((\log T)^{-1})$, and the local factor cancels the apparent exterior $h_0^{1/2+\sigma}$, leaving $h_0^{-1/2+\varepsilon}$. Stirling gives $X_{\lambda,\gamma}(t) \ll T^{\sigma+\varepsilon}$; the other exact X factor has size $T^{O((\log T)^{-1})}$. Standard vertical-strip zeta bounds then give (9.9), with the possible distance to $y = 0$ absorbed by a fixed power of $\log T$.

In the remaining tube $|y| < 1/(2\log T)$, apply the joint removability of the *complete* permutation sum from the logically prior Proposition 9.3 before taking absolute values. With γ fixed, write $\beta = -\gamma - w/\log T$. The maximum principle on $|w| \le 1/2$ bounds every value in the tube by the already established bound on $|w| = 1/2$. That circle changes only the logarithmic factor, and x, z remain uniformly separated from zero. This proves (9.9) throughout the tube, including the collision, without estimating either meromorphic summand there. The vertical-strip zeta bounds and the uniform form of Stirling's formula used above hold for every $\Im u$; the complementary compact ranges of the gamma arguments are bounded directly. Thus (9.9) holds on the complete vertical line, where $\Gamma(u)$ dominates the displayed polynomial in $|\Im u|$. $\square$

Set $\sigma = M + 1/2$ and move the common u-line in (8.17) from c_u to $-\sigma$. The complete residue list is as follows.

(i) At $u = 0$, $\Gamma(u)$ has residue one and $Q(0) = 1$; this gives $\mathcal{P}_\alpha$.
(ii) The gamma residues at $u = -j$, $1 \le j \le M$, vanish because $R_M(-j) = 0$.
(iii) The arithmetic collision locations $u = -\alpha - \gamma$ and $u = \beta - \alpha$ are removable in the complete $\mathcal{P}$ by the logically prior Proposition 9.3.
(iv) The poles $u = 1/2 + 2n - \alpha - it$ of $X_{\alpha+u,\gamma}(t)$ lie to the right of c_u and are not crossed. The $n = 0$ pole encountered in the preliminary move from $\Re u = 1$ to c_u is exponentially small. Denominator gamma factors give zeros.

There are no other poles between the two lines. By Lemma 9.1, the new line for fixed h, k is bounded by

$$T(\log T)^{C_M}(hk)^\varepsilon \left\{(T/Y)^\sigma + (h/Y)^\sigma\right\}. \tag{9.10}$$

After the outer sum, choose M so that

$$(B_0 - 1)\sigma > \frac{1}{2}(1 - \eta) + D + 1 + 2\varepsilon. \tag{9.11}$$

Then (9.10) is $O(T^{-D})$ after the outer sum.

9.3. **Joint removability at shift collisions.** The next lemma and proposition are grouped here because they describe the collision geometry used in the final contour moves. Logically, however, they precede Proposition 8.2. The lemma is formal local algebra. The proposition uses only (8.1), Euler-factor symmetry, and $X_{-\gamma,\gamma} = 1$; the separate fixed-s application also uses the residue pairings (8.7)–(8.9). None of these inputs imports quantitative continuation from Proposition 8.2. This is the forward dependency recorded explicitly just before that proposition.

Lemma 9.2 (Three-divisor cancellation)**.** *Let x, y be independent local holomorphic coordinates and put $z = x + y$. Suppose a, b, c, H_0 are holomorphic and*

$$F = \frac{a}{xz} + \frac{b}{yz} + \frac{c}{xy} + H_0, \qquad a + c \in (x), \quad b + c \in (y), \quad a - b \in (z). \tag{9.12}$$

Then F is holomorphic across $xyz = 0$.

Proof. Over the common denominator, the numerator is

$$N = ay + bx + cz. \tag{9.13}$$

Modulo x, use $z = y$ to obtain $N = (a + c)y$; modulo y, use $z = x$ to obtain $N = (b + c)x$; modulo z, use $y = -x$ to obtain $N = (b - a)x$. Thus each of the pairwise coprime linear factors x, y, z divides N, and therefore $xyz \mid N$ in the local analytic ring. □

For the fixed-s assembly, take the coordinates in (8.12). After extracting the simple polar zeta factors, the direct diagonal, reflected dual diagonal, and zero mode have denominator patterns xz, yz, and xy. The three residue cancellations in (8.7)–(8.9) give exactly the three divisibilities in (9.12). Thus the complete fixed-s integrand is jointly holomorphic even at double or triple intersections. Its separate $1/s$ factor remains, so the prescribed $s = 0$ residue is retained.

For the permutation sum, the collision pairs are

hyperplane	first Euler product	matching Euler product
$\lambda = -\gamma$	$Z_{-\gamma,\beta,\gamma}$	$Z_{-\gamma,\beta,\gamma}$, $X_{-\gamma,\gamma} = 1$
$\lambda = \beta$	$Z_{-\gamma,\beta,-\beta}$	$Z_{\beta,-\gamma,-\beta}$, equal by first-shift symmetry
$\beta = -\gamma$	$Z_{\lambda,-\gamma,\gamma}$	$Z_{\lambda,-\gamma,\gamma}$, $X_{-\gamma,\gamma} = 1$

In each row the polar zeta arguments are opposite local coordinates, so the residues have opposite signs, including the finite factors at primes dividing hk.

Proposition 9.3. *The complete expression $\mathcal{P}_\lambda(h, k; t)$ in* (8.1) *is jointly holomorphic across all shift-collision divisors.*

Proof. Apply Lemma 9.2 with

$$x = \lambda + \gamma, \qquad y = -\beta - \gamma, \qquad z = \lambda - \beta = x + y. \tag{9.14}$$

The middle, last, and first terms of (8.1) supply the denominator patterns xz, yz, and xy, respectively. The three rows of the preceding table are precisely the required divisibilities. This also handles the common collision locus $\lambda = \beta = -\gamma$, rather than only generic points of the individual hyperplanes. $\square$

9.4. **Proof of the main theorem.** By (3.13), the coupled smoothed expansion differs from (2.7) by an arbitrary negative power. Its $r = 0$ diagonals and fixed-orientation zero modes satisfy (8.21). The diagonal boundary is reduced by (9.8) and bounded by (9.7). The complete permutation line contributes only its $u = 0$ residue, by (9.10). Finally, the nonzero frequencies are bounded by (5.28). We obtain

$$\begin{aligned}\mathcal{I}_{\alpha,\beta,\gamma}(a,b) = &\sum_{h\le H}\sum_{k\le K}\frac{a_h\overline{b_k}}{\sqrt{hk}}\int_{\mathbb{R}} w(t)\mathcal{P}_\alpha(h,k;t)\,\mathrm{d}t\\ &+ O_{\eta,C,\varepsilon}\Big(T^\varepsilon\{KH^{3/2} + T^{1/2}(HK)^{1/2}\}\Big).\end{aligned}$$

By Proposition 9.3, the main term is jointly holomorphic at every shift collision. This is (2.11). The replacement (2.12) follows uniformly from Stirling's formula, completing the proof.

10. The limiting Bui corner

This section is a boundary calculation and is not used in the proof of Theorem 2.1. Write Bui's two mollifier lengths as

$$H \le T^{4/7-\varepsilon_B}, \qquad K \le T^{1/4-\varepsilon_B}. \tag{10.1}$$

Then

$$\frac{3}{2}\left(\frac{4}{7} - \varepsilon_B\right) + \left(\frac{1}{4} - \varepsilon_B\right) = \frac{31}{28} - \frac{5}{2}\varepsilon_B. \tag{10.2}$$

It is less than one precisely when

$$\varepsilon_B > \frac{3}{70}. \tag{10.3}$$

Thus Theorem 2.1 already covers every Bui box with this strict margin.

If the margin is suppressed and only the limiting exponents $(\theta_1, \theta_3) = (4/7, 1/4)$ are compared, the nonzero-mode estimate is

$$KH^{3/2} = T^{31/28+o(1)}. \tag{10.4}$$

To reach an error $T^{1-\delta}$ through the same reduction would require an additional saving of

$$T^{3/28+\delta} \tag{10.5}$$

in the limiting exponent budget. The pointwise completed Weil bound in Lemma 5.1 does not provide it for two arbitrary coefficient sequences.

An averaged reciprocal or Kloosterman-fraction estimate retaining both coefficient sequences could be a sufficient new input for the nonzero frequencies, but no such estimate is assumed or proved here. Nor would it by itself settle every analytic detail at the corner: the endpoint-deletion and full-kernel estimates in Section 6 were proved and audited under (2.8), and their uniformity at the limiting corner has not been established. It is therefore not valid to take a minimum over different Poisson orientations or to describe a single averaged estimate as the unique remaining dependency.

Finally, the cubic product in (2.7) is the "twisted third moment" appearing in Bui's mollifier cross term. It is not the sixth moment $\int |\zeta(1/2+it)|^6 \, \mathrm{d}t$, and Theorem 2.1 does not imply a sixth-moment asymptotic.

Appendix A. Proof ledger and reproducibility

This appendix records the dependency structure used to check the proof. It is not a substitute for the arguments in the body of the paper.

A.1. **Dependency ledger.**

ID	claim	location
A1	The unnormalized AFE includes all four remote completed-zeta residues, and the long zeta series is expanded only on $\Re u = 1$.	Lemma 3.1, (3.9)
A2	The coupled weight has simultaneous product decay on complete vertical contours in all four size quadrants.	Lemma 3.2
A3	Extracting $d = (h, km)$ gives a coprime inverse and exact Poisson density d/h.	(4.4)–(4.6)
A4	The fixed-l nonzero frequencies are $O(T^{\varepsilon} KH^{3/2})$, with the gcd factor in both completion terms averaged by (5.5) and the actual TS Poisson-integral size retained.	Lemma 5.1 and propositions 5.2 and 5.3
A5	The artificial $l < 1$ and $n < 1$ endpoint regions have arbitrary saving on the good half-space; the singular layer has exact gcd cancellation.	Proposition 6.2, (6.22)
A6	All four branches reach genuine Tonelli domains. At finite height the source and destination satisfy the rectangle identity including both horizontal edges; compact-uniform dominators then justify, in order, $V \to \infty$, $R \to \infty$, and $\kappa \to 0^+$.	(6.45), (6.48), (6.50)
A7	The beta integrals, finite Euler product, and archimedean reflection identities give the evaluated zero modes.	Section 7
A8	Three pairs of fixed-u small s-residues cancel before the diagonal boundary is separated.	(8.7)–(8.9)
A9	Quantitative continuation is applied only to $\mathscr{A}_T - \mathscr{R}_T - \mathscr{D}_T$.	Proposition 8.2
A10	The diagonal-boundary and permutation u-moves retain only their $u = 0$ residues; $-1, \ldots, -M$ are killed by R_M.	Section 9
A11	Double and triple shift collisions are jointly removable by local division by three linear factors.	Lemma 9.2 and proposition 9.3
A12	Combining the zero mode, nonzero mode and diagonal boundary gives the error stated in Theorem 2.1.	final paragraph of Section 9

The expository numbering does not reverse the proof dependency between A9 and A11. The collision statement A11 is proved directly from (8.1), (2.5), (2.6), and local three-divisor division; it is then used in the polynomial majorant inside A9. Thus the directed edge is A11 $\to$ A9, and there is no edge from A9 back to A11.

A.2. **Additional adversarial checks.** We record eight stress tests applied to the quantitative and uniformity claims used in the proof.

D1. The coefficient hypotheses are used through the actual block norms $\mathcal{A}_{H_0}$ and $\mathcal{B}_{K_0}$ in (5.6), or through the explicit outer absolute sum in

(8.31). After choosing a smaller auxiliary exponent,

$$\sum_{h\le H}\sum_{k\le K}\frac{|a_h b_k|}{\sqrt{hk}} \ll_\varepsilon T^\varepsilon (HK)^{1/2}.$$

No ℓ^1 coefficient norm is replaced by a supremum without its associated counting factor.

D2. Every occurrence of T^ε is interpreted after choosing the auxiliary divisor and localization exponents sufficiently smaller than the final ε. In particular, (5.19) is obtained only after the fixed completion order is known. The logarithmic dyadic counts and the finitely many subsequent losses may therefore be combined into one final T^ε. The coarser absorption used to prove the polynomial majorant in Proposition 8.2 is not used as part of the sharp error term. The corresponding implicit-constant and weight-seminorm dependencies are exactly those recorded in (2.2)–(2.3); in particular, fixed M permits $(\log T)^{C_M} \ll_{M,\varepsilon} T^\varepsilon$ but not the absorption of any fixed $T^{\delta(M)}$.

D3. The only size consequences used beyond (2.8) are

$$HK \le T^{1-\eta}, \qquad H \le T^{2(1-\eta)/3}, \qquad K \le T^{1-\eta},$$

which follow from $H, K \ge 1$. These are sufficient for (6.7), (6.9), the polynomial majorants, and the two final left-line estimates. No ordering between H and K, or stronger hidden length condition, is used.

D4. All differentiation and integration-by-parts orders are fixed before T. Finite completion uses only a fixed number of derivatives, while the endpoint orders depend on the prescribed fixed saving. The parameter order

$$(B_0, C_0, D) \longrightarrow M \longrightarrow N \longrightarrow (J, P, Q)$$

is recorded in (8.40)–(8.41). Consequently only finitely many seminorms from (2.1) enter the implied constants.

D5. After scaling the shifts by $\log T$, the domains (8.25) and their enlargement have a fixed collar, and the contours in (8.24) are independent of T and of the point in the target compact set. The threshold (8.28) keeps the kernel divisors and the relevant moving poles a fixed distance from these contours. Thus no continuation argument relies on a collar shrinking with the shifts.

D6. Uniformity is checked simultaneously at

$$\alpha + \gamma = 0, \qquad \beta + \gamma = 0, \qquad \alpha - \beta = 0,$$

including the common locus $\alpha = \beta = -\gamma$. The local coordinates in (9.14) satisfy $z = x + y$, and Lemma 9.2 proves divisibility by the full product xyz, rather than cancellation only at generic points. Fixed-radius Cauchy estimates on the scaled domain preserve the stated uniform bounds at the filled-in values.

D7. Individual meromorphic terms are not estimated on a collision divisor and then combined formally. The complete permutation sum is retained before (8.34) is applied; the direct diagonal, reflected dual diagonal, and zero mode are assembled before the boundary estimate (8.37); and the source object is bounded from its exact-cutoff representation. Cancellation therefore precedes every absolute-value estimate that crosses a collision.

D8. The final error ledger contains both surviving contributions: (5.28) gives $T^{\varepsilon}KH^{3/2}$, and (9.7) gives $T^{1/2+\varepsilon}(HK)^{1/2}$. The AFE recovery error, the continued endpoint error, and both regulator left lines are smaller than any prescribed fixed power by (8.21), (9.8), and (9.10). Thus the terms retained in (2.11) cover the complete error decomposition.

Scope of the audit. No first unsupported inference was found in these eight internal checks. This is a line-by-line consistency audit, not a formal verification or an external peer-review certificate. The finite numerical tests likewise check only explicit identities. Independent expert review remains necessary, especially for the regulator–Tonelli bridge and the assembled continuation and residue arguments.

A.3. **Finite identity checks.** The accompanying script `verification/check_zero_identities.py` uses arbitrary-precision arithmetic to test the two gamma identities and the linear approach to two collision divisors. A reference run at 80 decimal digits produced relative errors between approximately 10^{-52} and 10^{-80} for the gamma identities. In the collision tests, decreasing the displacement from 10^{-12} to 10^{-16} to 10^{-20} decreased the successive differences by a factor of 10^{-4} at each step. These are finite consistency checks only; the contour deformations, Tonelli arguments, and continuation are proved analytically in the body of the paper.

A.4. **Release integrity.** The source archive distributed with this paper contains the complete manuscript source, bibliography, compiled PDF, numerical verification script, and its fixed-version arbitrary-precision dependency and license. The file `MANIFEST.sha256` records a SHA256 digest for every other payload file in the archive. The verification script uses explicit release thresholds and prints `status=PASS` only after all checks have completed.

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